Ain Shams University
Faculty of Computer & Information Sciences
Computer Science Department

Elevare: A Platform for Enhancing Public Speaking and Soft Skills

This documentation is submitted as required for a bachelor's degree in computer and information sciences.

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Cairo
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Abstract

Effective public speaking is a critical competency in academic and professional contexts, yet a large majority of individuals experience significant speech anxiety and lack access to structured, personalized coaching. Current AI-driven solutions address this challenge only partially, often focusing on isolated modalities (e.g., speech rate or facial expressions) while failing to deliver integrated, content-aware feedback grounded in the speaker's specific material. To bridge this gap, this paper presents Elevare, an advanced, AI-powered multimodal public speaking coach that unifies emotion recognition, presentation analysis, dynamic question generation, answer validation, and feedback synthesis into a single, cohesive framework.

The architecture of Elevare comprises three interconnected modules designed to provide a comprehensive training experience. The first module focuses on multimodal emotion recognition by integrating three independent deep learning classifiers to analyze core communication channels. Specifically, a Stacked Bidirectional LSTM network achieves 92.9% accuracy in Speech Emotion Recognition across a five-dataset corpus, while an optimized GiMeFive convolutional architecture yields 84.46% accuracy for Facial Emotion Recognition on the RAF-DB dataset, and a Temporal Convolutional Network (TCN) captures body language emotional cues with 89.5% accuracy. The second module ensures contextual material mastery by utilizing a Semantic Graph Conversational Question Generation (SG-CQG) pipeline to transform user-uploaded presentation slides into contextually grounded practice questions. User responses to these questions are subsequently evaluated using a zero-shot Retrieval-Augmented Generation (RAG) framework powered by Llama 3.3-70b, which achieves a grading accuracy of 72.48%.

Finally, the third module implements a Dual-Agent Feedback Architecture that effectively decouples objective analytical reasoning from empathetic coaching language. Powered by Gemini 2.5 Flash, the first agent generates structured, bias-controlled performance assessments, while a secondary agent synthesizes the multimodal outputs into personalized, actionable feedback reports. By shifting automated presentation tracking into a holistic, interactive, and content-aware coaching experience, Elevare empowers users to master both their physical delivery and the contextual depth of their material.

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List of Abbreviations

Abbreviation	Stands for
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
AQG	Automated Question Generation
ASAS-F	Automatic Short Answer Scoring with Feedback
ASAS-F-Z	Zero-Shot Automatic Short Answer Scoring with Feedback
Bi-LSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CoQA	Conversational Question Answering
CQG	Conversational Question Generation
CTA	Call To Action
DEB	Detail Extraction Block
DSPy	Declarative Self-improving Python
DRW	Deferred Re-Weighting
FER	Facial Emotion Recognition
FERPlus	Facial Expression Recognition Plus
GUI	Graphical User Interface
LDA	Linear Discriminant Analysis
LDAM	Label-Distribution-Aware Margin
LLM	Large Language Model
LMS	Log Mel Spectrogram
MFCC	Mel-Frequency Cepstral Coefficient
MMLA	Multimodal Learning Analytics
OCR	Optical Character Recognition
PCA	Principal Component Analysis
Q&A	Question and Answer
QAV	Question Answer Validation
QG	Question Generation
RAG	Retrieval-Augmented Generation
RAF-DB	Real-world Affective Faces Database
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
SER	Speech Emotion Recognition
SGD	Stochastic Gradient Descent
STFT	Short-Time Fourier Transform
TCN	Temporal Convolutional Network
UI	User Interface

VR	Virtual Reality
ZCR	Zero Crossing Rate

Chapter 1
Introduction

Chapter 1: Introduction

1.1 Motivation

The possession of strong technical knowledge and well-developed ideas is widely regarded as the cornerstone of academic and professional success; yet this foundation is rendered significantly less effective when individuals are unable to articulate it with clarity and confidence. In many cases, the barrier is not a lack of competence but a lack of communication — and nowhere is this more apparent than in public speaking and presentation contexts. Empirical evidence reflects the scale of this challenge: approximately 50% of students experience high levels of public speaking anxiety, 42% moderate levels, and only 9% low levels [1], as illustrated in Figure 1.1, while in professional contexts, candidates consistently lose career opportunities not due to insufficient technical expertise but due to ineffective verbal and nonverbal delivery [2–4]. The need for effective, accessible communication training is therefore both urgent and widespread.

What makes this moment particularly significant is that recent technological advances have, for the first time, made a comprehensive solution genuinely achievable. The rapid maturation of large language models has enabled context-aware natural language understanding at scale; advances in computer vision have made real-time gesture and facial expression analysis both accurate and computationally feasible; and transformer-based speech models now permit nuanced vocal analysis that captures not just what is said, but how it is delivered [5, 6]. Together, these developments create a convergence point at which a unified, multimodal training platform is no longer a research ambition but a practical engineering goal. Yet despite this technological readiness, no existing tool has brought these capabilities together into a single, content-aware training experience — and that is the gap this project addresses.

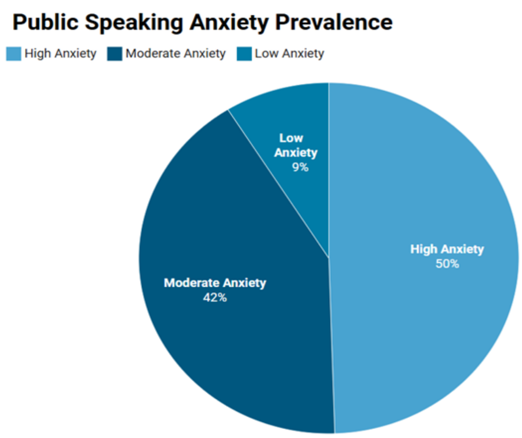

Figure 1. 1 Distribution of public speaking anxiety levels among university students [1].

1.2 Problem Definition

The specific problem this project addresses is the absence of a unified, content-aware public speaking training system. While individual tools exist for delivery assessment or question practice, they address these needs in isolation and share three critical limitations. First, delivery feedback is generic — evaluating surface metrics such as pace, volume, or filler word frequency without connecting them to the specific content being presented or the speaker's recurring weaknesses [5]. Second, practice questions are content-agnostic, generated from fixed templates rather than derived from the user's actual presentation material, making them poor preparation for real high-stakes scenarios [8]. Third, the cognitive level and background of the intended audience are entirely overlooked — knowing who you are speaking to is a fundamental dimension of communication that current tools fail to address [6].

These gaps are particularly consequential for two groups: university students preparing for academic presentations, and early-career professionals facing interviews or client-facing roles [4, 7]. Both groups require not just generic practice, but a structured, personalized training experience grounded in their own content [8]. The problem this project solves is therefore precisely defined: the lack of an integrated platform that combines multimodal performance analysis, content-derived question generation tailored to audience personas, and personalized feedback — all within a single, accessible system.

1.3 Objective

The main objective of this project is to develop an AI-powered application "Elevare" that improves communication and fosters self-expression in presentations. The following specific objectives were defined to achieve this goal.

1. Develop an AI-driven feedback system that analyzes verbal and nonverbal communication cues to generate specific, actionable coaching tips — such as pacing, filler-word usage, and eye contact — aimed at helping users identify and improve individual aspects of their public speaking skills.
2. Build an integrated evaluation system that fuses a user's body language, vocal variety, and facial expressions into a single, unified performance score, providing a holistic assessment of overall presentation delivery.
3. Develop an automatic question generation and validation system that reads the user's presentation content, such as uploaded slides, to generate topic-specific practice questions and evaluate user responses for semantic correctness and relevance, enhancing practice sessions with realistic challenges.

1.4 Time Plan

The project was executed across an academic year spanning from October 2025 to June 2026. The work plan covers all major phases from initial research through implementation, integration, testing, and final delivery. Table 1.1 summarizes the planned timeline for each activity.

Table 1. 1 Time Plan Summary.

Project Activities	Start Date	End Date
Survey on existing studies	29/09/2025	30/10/2025
Data Collecting and preprocessing	20/10/2025	10/11/2025
Implementation (Phase 1)	11/11/2025	22/12/2025
Final Exams	12/11/2025	21/12/2025
Implementation (Phase 2)	12/01/2026	31/03/2026
Build GUI	12/01/2026	15/04/2026
Whole System integration and testing	12/01/2026	30/04/2026
Maintain System	01/04/2026	31/05/2026
Writing Documentation	20/10/2025	30/06/2026
Closing and handing over of the project	01/06/2026	30/06/2026

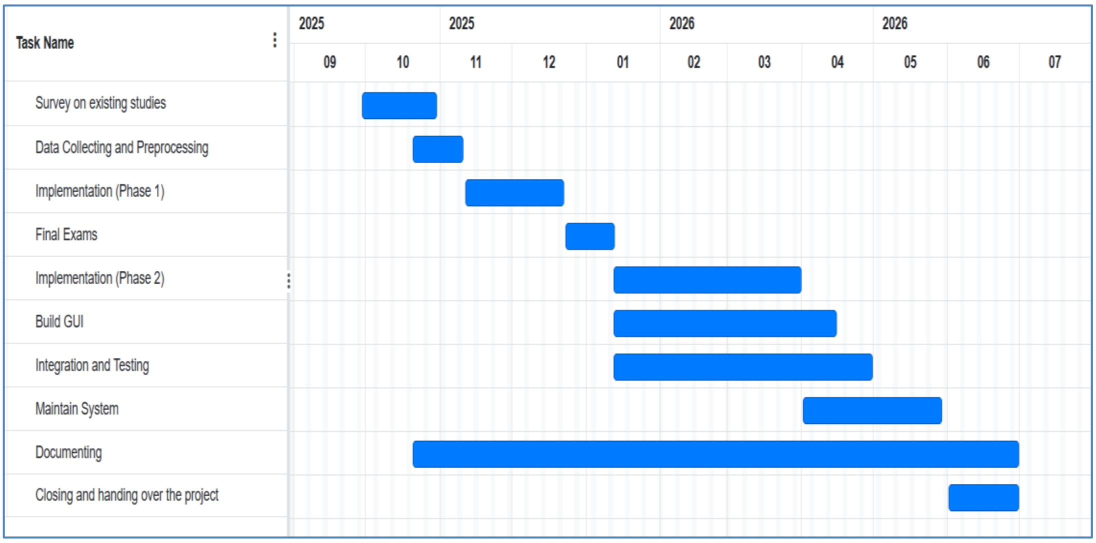

Figure 1. 2 Time Plan Gantt Chart.

1.5 Document Organization

The remainder of this document is organized as follows.

Chapter 2 (Background) presents a comprehensive review of the relevant literature and prior work in the areas of public speaking anxiety, multimodal emotion recognition, automatic short answer scoring, and AI-driven feedback systems. It also surveys existing tools and systems similar to Elevare, providing the necessary context for the design decisions made throughout this project.

Chapter 3 (Analysis and Design) describe the overall system architecture of Elevare and the relationships between its core modules. It covers the intended users of the system and their characteristics, and includes the system overview, use case diagram, class diagram, sequence diagram, and database diagram that collectively model the system's structure and behavior.

Chapter 4 (Implementation and Testing) provide a detailed description of all implemented modules, including Facial Emotion Recognition, Speech Emotion Recognition, Body Language Emotion Recognition, Question Generation, and Question Answer Validation. For each module, the datasets, preprocessing steps, model architectures, and design decisions are documented. The chapter further covers the UI design and wireframes, the testing procedures applied across the system, and a User Manual with operating instructions.

Chapter 5 (Conclusion and Future Work) summarize the contributions of the project, reflects on the outcomes achieved, and outlines potential directions for future improvement and extension of the system.

Chapter 2
Background

Chapter 2: Background

To establish a robust foundation for the proposed system, a comprehensive investigation of prior research and existing AI-driven solutions for presentation skill enhancement was conducted. Analyzing these methodologies provided critical insights into established frameworks, widely adopted datasets, and the baseline performance of diverse artificial intelligence models. Accordingly, this chapter systematically reviews the literature relevant to the core dimensions of this project.

The scope of this review encompasses state-of-the-art studies in facial emotion recognition, speech emotion recognition, body language analysis, automatic question generation, question-answering systems, and AI-based feedback synthesis leveraging deep learning and natural language processing (NLP) techniques. For each reviewed study, the utilized datasets, preprocessing methodologies, architectural models, and evaluation metrics are critically examined. Furthermore, this chapter outlines the general architectures governing current AI coaching and behavioral analysis platforms, concluding with a comparative analysis that delineates the unique contributions and performance improvements of the proposed system over existing benchmarks

2.1 General Overview

Presentation skills play a vital role in academic, professional, and social environments, as they directly influence the effectiveness of communication and audience engagement. Traditional methods for improving presentation skills often rely on human evaluation and manual feedback, which may be subjective, time-consuming, and not always accessible to all users. With the rapid advancement of artificial intelligence technologies, researchers have explored the use of AI-based systems to automatically analyze and enhance public speaking and presentation performance.

Recent developments in computer vision, speech processing, and natural language processing have enabled the creation of intelligent systems capable of evaluating multiple aspects of human presentations. These systems can analyze facial expressions, speech emotions, body posture, voice characteristics, and verbal content to provide personalized feedback and performance assessment. In addition, modern AI models can generate audience-like questions, evaluate answers, and offer constructive suggestions for improvement, creating a more interactive and effective training experience.

Most existing systems focus on a limited number of presentation-related features, such as speech analysis or pronunciation assessment, while fewer systems combine multiple modalities into a unified intelligent coaching platform. Therefore, there is a growing need for comprehensive AI-based presentation coaching systems that integrate facial emotion recognition, speech analysis, body language evaluation, question generation and answering, and automated feedback generation within a single framework.

This chapter reviews the most relevant studies and technologies related to the modules used in our proposed system. The reviewed literature includes approaches based on deep learning, computer vision, audio signal processing, and transformer-based natural language processing models. We also discuss datasets, preprocessing techniques, evaluation metrics, and architectures commonly used in these systems. Finally, a comparative analysis is presented to highlight the limitations of existing solutions and demonstrate the unique contributions of our proposed AI presentation coach system.

While existing AI-driven coaching platforms tend to address individual modalities in isolation, a fully integrated system would follow a common architectural pattern. Such a pipeline would begin with preprocessing, encompassing silence filtering, facial alignment, skeletal coordinate normalization, and text chunking across all input streams. A feature extraction stage would subsequently derive modality-specific representations such as Mel-Frequency Cepstral Coefficients (MFCCs) for speech and joint angles for body posture. A classification and generation stage would then interpret these features by assigning affective states (e.g., confident, nervous, neutral) and producing context-aware question–answer pairs grounded in the presentation content.

Finally, a feedback synthesis layer would consolidate these outputs into a unified architecture capable of delivering both objective performance scoring and actionable pedagogical guidance. The proposed system, Elevare, is designed around this complete pipeline, directly addressing the modality and functional gaps identified in existing tools.

2.2 Related Work

Based on computer vision, audio processing, and multimodal deep learning, researchers have actively developed models to automate human behavior analysis, emotional state recognition, and communication skill assessment from video and audio streams. Numerous studies have explored various architectures and datasets for building robust platforms in the fields of human-computer interaction, educational technology, and public speaking analytics.

2.2.1 Facial Emotion Module

Bingyu Nan et al. [9] developed a visual self-attention mechanism network for facial expression recognition. The system was trained on the RAF-DB dataset, which contains 30,000 facial images distributed across 7 emotion classes. The data underwent standard preprocessing steps that included face detection using RetinaFace, face alignment, and data augmentation through random horizontal flipping. The model employed a Truncated ConvNeXt architecture paired with a Detail Extraction Block, achieving a high accuracy of 97%. The paper emphasized the effectiveness of self-attention mechanisms in isolating subtle facial micro-expressions.

J. Wang and L. Kawka [10] proposed the GiMeFive architecture, a lightweight and interpretable model tailored for facial emotion classification. The research utilized aligned and cropped images from the RAF-DB dataset, which encompasses approximately 10,000 images distributed across 6 primary emotion classes. Preprocessing steps were kept minimal, relying on image resizing to a 64×64-pixel resolution and standard normalization without the application of random spatial augmentations. By employing a compact convolutional design equipped with batch normalization and dropout and trained using standard cross-entropy loss with stochastic gradient descent, the model achieved a baseline test accuracy of 86.5%. This study highlighted the efficiency of streamlined network architectures in accurately identifying facial expressions while maintaining high computational simplicity.

Di Chang et al. [11] introduced LibreFace, an open-source toolkit for deep facial expression analysis designed to support multiple facial behavior understanding tasks, including facial expression recognition, action unit detection, valence-arousal estimation, and facial attribute analysis. The framework integrates several state-of-the-art deep learning architectures and provides unified pipelines for preprocessing, training, evaluation, and real-time inference. The study utilized widely adopted facial expression datasets such as RAF-DB, AffectNet, and Aff-Wild2, while preprocessing procedures included face detection, facial alignment, normalization, and data augmentation techniques. LibreFace supports both convolutional neural networks and transformer-based architectures, enabling efficient experimentation and deployment for affective computing applications. Experimental evaluations demonstrated strong performance across multiple benchmark datasets, achieving facial expression recognition accuracies of 90.67% on RAF-DB and 64.96% on AffectNet, while also obtaining competitive results in valence-arousal estimation and action unit detection tasks. The study highlighted the effectiveness of LibreFace as a flexible and reproducible platform for facial emotion analysis and multimodal human behavior understanding.

2.2.2 Speech Emotion Module

Shahana Yasmin et al. [12] introduced a deep learning technique for Speech Emotion Detection using feature engineering. The study used a combined dataset composed of RAVDESS, TESS, SAVEE, EmoDB, and CREMA-D, comprising 12,424 audio-visual samples across 7 emotion classes. Preprocessing steps included silence removal for segments exceeding 200ms, audio resampling to 22,050 Hz, and data augmentation techniques such as Gaussian noise injection, pitch shifting, and time stretching. Their model utilized a Deep Conditional Random Field stacked with a Bidirectional LSTM (DCRF-BiLSTM), achieving an accuracy of 94.04%. This hybrid architecture effectively captures complex acoustic properties, offering a robust foundation for vocal emotion recognition.

Akinpelu, Viriri, and Adegun [13] proposed an enhanced Speech Emotion Recognition (SER) system using a lightweight Vision Transformer (ViT) architecture. The study utilized the TESS and EMO-DB benchmark datasets, where speech signals were preprocessed through noise reduction, normalization, and mel-spectrogram extraction to transform audio signals into image-like representations. The proposed ViT-based model leveraged self-attention mechanisms to capture spatial and emotional patterns from spectrogram inputs more effectively than traditional CNN-based methods. Experimental results demonstrated strong performance, achieving accuracy of 98% on TESS, 91% on EMO-DB, and 93% on combined datasets, highlighting the effectiveness of transformer-based architectures for speech emotion recognition tasks.

Akinpelu, Viriri, and Haroon Yousaf [14] introduced SwinTSER, a bilingual Speech Emotion Recognition (SER) framework based on the Shifted Window Transformer (Swin Transformer) architecture. The system was designed to improve emotion recognition performance across multilingual speech datasets by capturing both local and global emotional dependencies within speech signals. The proposed approach utilized hierarchical transformer layers with shifted window attention mechanisms to efficiently model temporal and spectral emotional features while reducing computational complexity. The study employed bilingual speech emotion datasets with preprocessing steps that included noise reduction, normalization, and mel-spectrogram extraction for transforming audio signals into image-like representations suitable for transformer-based learning. Experimental evaluations demonstrated strong performance, achieving recognition accuracies of 98.21% on TESS, 93.75% on EMO-DB, and 95.43% on combined bilingual datasets, outperforming several existing CNN- and transformer-based SER approaches. The study emphasized the effectiveness of Swin Transformer architectures for multilingual and real-time speech emotion recognition applications.

2.2.3 Body Language Module

Vaijayanthi Sekar and J. Arunnehru [15] explored human emotion recognition from body posture using multiple machine learning techniques. The study utilized the GEMEP dataset, which contains 145 video recordings across 17 emotion classes. Preprocessing included video-to-frame extraction, skeleton pose detection, and coordinate normalization between 0 and 1. Several machine learning models were evaluated, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Naïve Bayes. Among them, the KNN model achieved the highest performance, reaching an accuracy of 97.1%, demonstrating the effectiveness of kinematic body posture features for emotion recognition.

P. Müller et al. [16] proposed an approach for recognizing emotion regulation strategies from human behavior using Large Language Models (LLMs). The study utilized the DEEP corpus, which contains multimodal behavioral data and annotations for seven emotion regulation classes in social interaction scenarios. The system incorporated verbal and nonverbal behavioral cues, person-specific factors, and contextual information to generate prompts for fine-tuning Llama2-7B and Gemma models using Low-Rank Adaptation (LoRA). Multiple configurations were evaluated, and the fine-tuned Llama2-7B model achieved the best performance with an accuracy of 84%, outperforming previous Bayesian Network–based approaches. The study demonstrated the effectiveness of combining multimodal behavioral analysis with LLMs for understanding complex human emotional behavior.

T. Wang et al. [17] proposed a multiscale spatio-temporal network for emotion recognition from full-body motion. The study utilized five public body-motion emotion datasets, including EGBM, KDAE, Emilya, MPI, and DMCD, containing skeletal motion sequences across multiple emotional categories. Preprocessing involved extracting and normalizing 3D skeletal joint coordinates and selecting emotionally informative temporal segments using a pseudo-energy–based scale selection algorithm. The proposed framework employed multiscale spatio-temporal convolutional neural networks (CNNs) and hierarchical fusion mechanisms to capture both coarse-grained body movements and fine-grained posture variations across time. In addition, the model integrated posture covariance matrices and 3D posture image representations to jointly learn spatial and temporal emotional features. Experimental results demonstrated that the proposed method achieved average classification accuracies exceeding 86% across all evaluated datasets and outperformed several state-of-the-art body-motion emotion recognition approaches, highlighting the effectiveness of multiscale spatio-temporal feature learning for affective body expression analysis.

Chiara Plizzari, Marco Cannici, and Matteo Matteucci [18] proposed a skeleton-based action recognition framework using Spatial and Temporal Transformer Networks (ST-TR). The study utilized large-scale skeleton datasets, including NTU-RGB+D 60, NTU-RGB+D 120, and Kinetics Skeleton 400. The proposed architecture employed Spatial Self-Attention (SSA) to model relationships between body joints within a frame and Temporal Self-Attention (TSA) to capture motion dependencies across frames. Experimental evaluations demonstrated state-of-the-art performance, achieving accuracies of 89.9% and 95.3% on NTU-RGB+D 60 under cross-subject and cross-view evaluation settings, respectively, while also obtaining competitive results on NTU-RGB+D 120 and Kinetics Skeleton 400. These results highlighted the effectiveness of transformer-based architectures in understanding complex body movement patterns. Although the study focused on action recognition rather than emotion recognition, the proposed ST-TR framework is highly relevant to body language emotion analysis, as emotional states can also be inferred from spatial and temporal skeletal movement patterns.

C. Beyan et al. [19] proposed a full-body emotion classification framework based on modeling multiple temporal scales of human movement. The study utilized the GEMEP corpus, which contains acted emotional body expression sequences covering multiple emotion categories. Preprocessing involved extracting full-body skeletal motion features and segmenting motion sequences into different temporal scales to capture both short-term and long-term expressive dynamics. The proposed framework employed multiscale temporal modeling combined with handcrafted kinematic descriptors to analyze body movement patterns associated with emotional behavior. Experimental evaluations demonstrated that the multiscale approach achieved an accuracy of 81.3%, outperforming conventional single-scale motion analysis methods. The study highlighted the importance of modeling emotional body expressions across multiple temporal resolutions for improving full-body emotion recognition performance.

2.2.4 Question Generation Module

Xuan Long Do et al. [20] proposed SG-CQG, a two-stage framework for answer-unaware Conversational Question Generation (CQG) that addresses the challenges of what-to-ask and how-to-ask. The study utilized the CoQA dataset, a large-scale conversational question-answering dataset comprising 127,000 question-answer pairs derived from diverse referential contexts, with approximately 20% of questions being boolean. The what-to-ask module constructs a semantic graph from the context to select rationale sentences in a non-sequential, semantically coherent manner, while the how-to-ask module generates questions guided by explicit question type control signals and applies filtering heuristics to remove low-quality outputs. Experimental results demonstrated that SG-CQG achieved state-of-the-art performance on CoQA, attaining a BERTScore-entailment of 81.99, a Conv-Distinct score of 62.27, and a Context Coverage of 68.52%, outperforming all baselines including T5, BART, GPT-2, and CoHS-CQG, with human evaluation confirming superior factuality, conversational alignment, and answerability.

Subhankar Maity, Aniket Deroy, and Sudeshna Sarkar [21] proposed an automated question generation (AQG) framework for educational domains by leveraging In-Context Learning (ICL), Retrieval-Augmented Generation (RAG), and a novel Hybrid Model combining both approaches. The study utilized the EduProbe dataset, comprising 3,502 question-answer pairs drawn from NCERT textbook passages across History, Geography, Economics, Environmental Studies, and Science, spanning grades 6 through 12. Experimental results showed that ICL (k = 7) achieved the best automated scores with a ROUGE-L of 55.95 and BERTScore of 75.92, while the Hybrid Model (k = 5, m = 5) consistently outperformed all other approaches in human evaluation with scores of 4.84 for grammaticality, 4.74 for appropriateness, 4.25 for relevance, and 4.02 for complexity, demonstrating the pedagogical advantage of combining retrieval-based context enrichment with few-shot learning.

Nicy Scaria, Suma Dharani Chenna, and Deepak Subramani [22] proposed an automated educational question generation framework using five state-of-the-art LLMs: Mistral-7B, Llama-2-70B, Palm 2, GPT-3.5, and GPT-4 to generate questions across all six levels of Bloom's Taxonomy for a graduate-level data science course. Five prompting strategies of varying complexity were evaluated, ranging from simple zero-shot prompting to chain-of-thought prompting enriched with skill definitions and few-shot examples. Questions were assessed by two domain experts using a nine-item rubric covering understandability, grammaticality, relevance, answerability, and Bloom's level adherence. Results showed that question quality improved consistently as more information was added to the prompt, with GPT-4 using the most enriched prompt strategy achieving the highest quality and skill adherence scores, while automated LLM-based evaluation was found to be misaligned with expert human judgment across all models.

Vinay Samuel et al. [23] introduced PersonaGym, the first dynamic evaluation framework for assessing persona-based LLM agents, along with PersonaScore, an automated human-aligned metric grounded in decision theory. The framework was evaluated across 10 leading LLMs using a benchmark of 200 personas and 10,000 questions, revealing that increased model size and complexity do not necessarily lead to better persona adherence. Although the study does not focus on question generation, its persona-based interaction framework where LLMs assume specific personas to generate contextually aligned responses directly inspired the audience persona mechanism in our proposed system, which simulates different audience types to guide question generation style and cognitive level.

2.2.5 Question Answer Validation Module

Menna Fateen et al. [24] implemented a modular retrieval-augmented generation (RAG) system for automatic short answer scoring and feedback generation. The model was trained and evaluated on the saf_communication_networks_english dataset, comprising 300 rows of technical short-answer responses regarding communication networks. Utilizing a Zero-Shot learning approach powered by DSPy Chain of Thought prompting and the Llama 3.3-70b backbone, the model calculated strict classification labels and generated actionable feedback. The system achieved a peak accuracy of 72.48%, highlighting its capacity to provide objective, explainable coaching and precision scoring under zero-shot conditions.

Thuy Vu and Alessandro Moschitti [25] introduced AVA, an automatic evaluation framework designed to estimate the accuracy of question answering systems by measuring semantic alignment against gold-standard reference targets. The study leveraged multiple large-scale industrial and open-source benchmarks including WikiQA, Natural Questions, and TREC-QA compiling over two million evaluation triplets consisting of a question, a reference answer, and a system candidate response. Preprocessing steps involved transformer-based textual tokenization and sequential joint encoding of all three matrix components to preserve question-specific contextual boundaries. The underlying architecture utilized deep transformer language models configured to map high-dimensional sentence embeddings and compute semantic similarity scores. The framework achieved an F1 score of 74.7% in predicting human grading decisions and estimated overall QA system accuracy with a robust Root Mean Square Error (RMSE) between 0.02 and 0.09. This approach demonstrated the high reliability of automated, cross-reference language modeling pipelines in replacing expensive manual evaluation setups.

2.2.6 Feedback Generation Module

Soham Padia et al. [26] proposed a multimodal AI-driven framework aimed at enhancing public speaking skills through comprehensive analysis. The study utilized a distinct custom dataset: a Public Speaking Video Dataset comprising nearly 800 videos sourced from various online repositories and manual recordings. Preprocessing procedures involved video frame extraction, resizing, and speech transcription for textual analysis. The system integrated rule-based algorithms and multimodal tracking to dynamically evaluate verbal metrics such as speaking rate, pitch modulation, and filler word frequency alongside non-verbal cues like facial expressions. By synthesizing these cross-channel metrics into structured graphical analyses, the architecture successfully generated detailed, action-oriented feedback reports, providing users with a comprehensive summary of their behavioral strengths and targeted areas for improvement.

Alvaro Becerra et al. [27] introduced MOSAIC-F, a multimodal data-driven framework designed to enhance students' oral presentation skills through personalized feedback. The study evaluated real-world presentations using a dataset comprising 50 recorded sessions (incorporating both individual and group formats) delivered by 65 university students. Data collection and preprocessing involved synchronizing multiple complex sensor streams, including high-definition webcams, ambient microphones, eye-tracking glasses, and smartwatch physiological sensors. The framework integrated these objective Multimodal Learning Analytics (MMLA) with standardized, rubric-based collaborative assessments from peers and professors. By synthesizing these diverse streams, an AI-driven module generated automated, actionable feedback on nonverbal behaviors such as posture, speech patterns, and stress levels. The research demonstrated that combining human-based qualitative evaluations with objective sensor data provides a highly accurate, comprehensive feedback loop, significantly fostering student self-reflection and communication competency.

N. Basta et al. [28] presented DIALEVAL, a type-theoretic framework utilizing dual-agent architecture to automate the evaluation of instruction adherence and dialogue coherence. To eliminate systematic errors found in uniform LLM evaluations, the architecture deploys two specialized Claude-3.5-Sonnet agents: one agent strictly dedicated to extracting and decomposing typed predicates, and a second independent agent that performs type-specific satisfaction assessments based on human-aligned criteria. Evaluated across multiple LLM backbones, the dual-agent system achieved 90.38% accuracy and demonstrated a substantially stronger alignment with human judgment compared to single-agent state-of-the-art methods. This formally validates the strategy of functionally separating analytical reasoning tasks from final scoring mechanisms.

Lucas Jasper Jacobsen and Kira Elena Weber [29] investigated the efficacy of large language models as automated feedback providers in higher education environments. The study evaluated feedback generation aimed at teacher education, comparing AI-generated responses against traditional feedback provided by human novices and domain experts. To test the system, the researchers developed a theory-driven prompt engineering manual and designed three distinct prompt structures varying in pedagogical quality. Using ChatGPT-4 as the core generative model, the study demonstrated that when guided by the highest-quality structured prompt, the LLM not only outperformed novice feedback but also surpassed expert human feedback in key metrics such as explanation clarity, question generation, and specificity. The research underscores the necessity of precise prompt engineering and validates the use of LLMs as highly efficient, high-quality alternatives to traditional expert feedback mechanisms.

2.3 Comparative Analysis

Table 2.1 presents a brief comparison between the proposed Elevare framework, and the leading state-of-the-art commercial solutions related to AI-powered presentation coaching. The comparison evaluates each platform across four main criteria: primary focus, modalities evaluated, question and answer (Q&A) generation capabilities, and feedback architecture. While academic literature often focuses on isolated modalities, reviewing these active applications provides critical insight into current market capabilities and highlights the functional gaps that Elevare addresses.

Table 2. 1 Comparative analysis of state-of-the-art commercial AI speaking platforms.

Platform / System	Primary Focus	Modalities Evaluated	Q&A	Feedback Architecture
Orai	Speech delivery, communication improvement	Audio, Text, Gaze	None	metric-based automated feedback
Yoodli	speech delivery, pacing, clarity, filler words, and structured speaking feedback and Detection eye contact	Audio, Text, Basic Facial (Gaze)	None	Structured rubric-based communication feedback
VirtualSpeech	Immersive VR environmental simulation & soft skills	Audio, Text, Head Tracking (via VR)	Yes (Generative Roleplay)	Single-agent LLM & metric dashboards

- Unprecedented Multimodal Integration: Unlike existing systems (Orai and Yoodli) that focus primarily on speech delivery, text, and gaze tracking, *Elevare* introduces a more comprehensive assessment framework by incorporating body posture analysis alongside audio, facial, and text modalities. This fills a significant gap in evaluating physical delivery and body language.
- Context-Aware Dynamic Q&A: While platform-specific solutions like Orai and Yoodli lack any question-answering capabilities, and VirtualSpeech relies solely on generative roleplay, *Elevare* distinguishes itself by offering a context-aware Q&A module. This ensures that the simulated questions are

deeply grounded in the speaker's specific presentation content rather than general prompts.

- Advanced Dual-Agent Feedback Paradigm: *Elevare* evolves beyond traditional evaluation methods—which rely on rigid metric-based automated feedback (Orai), structured rubrics (Yoodli), or single-agent LLMs (VirtualSpeech). By implementing a Dual-Agent Architecture, the system successfully decouples cold, objective analytical metrics from empathetic coaching, offering a more nuanced, personalized, and human-like mentoring experience.
- Holistic Coaching vs. Isolated Tracking: Most commercial systems serve primarily as presentation tracking utilities. In contrast, *Elevare* represents a paradigm shift toward a holistic, interactive coaching ecosystem that targets both physical behavioral cues (facial expression, voice tone, and body posture) and conceptual mastery of the delivered

Chapter 3
System Analysis and Design

Chapter 3: System Analysis and Design

3.1 System Overview

In figure 3.1, a visualization of the full architecture of our project's system. The system is composed of three tiers. Presentation layer, application layer, and database layer. Over the next sections we shall go over them in more detail.

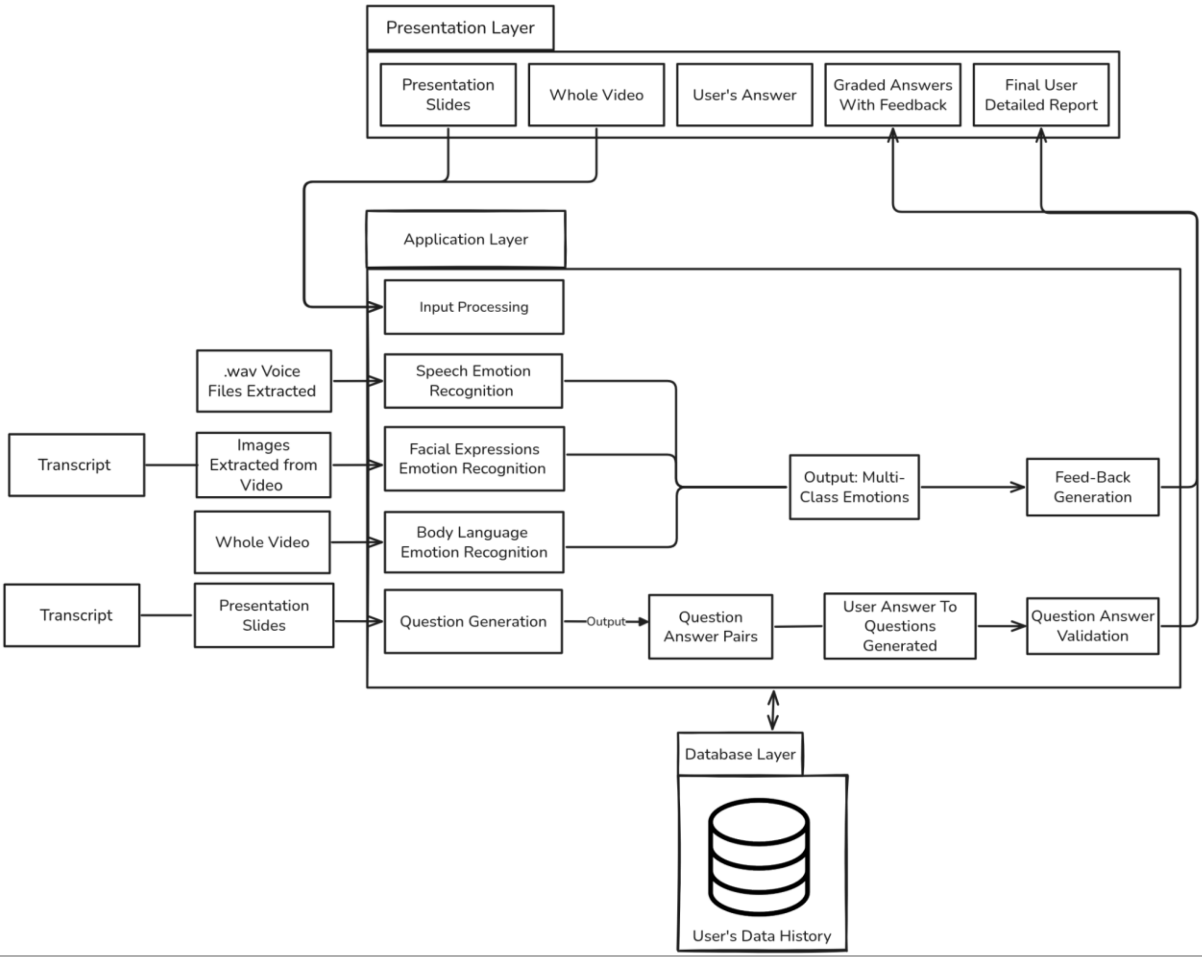

Figure 3. 1 Three Tier System Architecture.

Presentation Layer through this layer, the user starts to interact with the system by inserting the needed inputs: presentation slides, the presentation video of himself, and later his answer to the generated questions.

Application layer is the core computational engine of Elevare. It houses five distinct processing modules: the Speech Emotion Recognition Module, Facial Expressions Emotion Recognition Module, Body Language Emotion Recognition Module, Question Generation Module, and the Feedback Generation Module.

In this Data layer, the user data is preserved including his account and preferences, and the previous reports produced per previous presentations and practices.

3.2 Methods and Procedures

3.2.1 Data Ingestion & Preprocessing

First, the input data is digested and broken down into components needed for the rest of modules and to be processed to their use.

- Voice Preprocessing (For Speech Emotion Recognition - SER):
  Silence Removal: To isolate active speech and eliminate dead air, the system analyzes the audio and removes 70% of the silence from the beginning and end of each clip whenever the silence length exceeds 200ms [30].

  Resampling & Normalization: The truncated audio is resampled to a standardized 22050 Hz sampling rate to ensure uniform input dimensions. The data is subsequently normalized using Z-Score normalization [30].

  Data Augmentation: To ensure the model remains robust across different acoustic environments and to mitigate class imbalance, augmentation techniques are applied. These specifically include Gaussian noise injection, pitch shifting (altering frequency without changing duration), and time stretching (changing speed without altering pitch) [30].

- Facial Visual Preprocessing (For Facial Emotion Recognition-FER):
  Resizing & Channel Conversion: The video stream undergoes frame extraction. Each frame is scanned to detect the user's face (the region of interest), which is then cropped via a bounding box. The cropped facial image is resized to a standardized resolution, converted to grayscale while maintaining three channels, and normalized into a tensor array to match CNN input requirements [10].

- Skeletal Visual Preprocessing (For Body Language):
  The visual stream undergoes video-to-frame extraction with rate fitting all the modules for a proper processing parallelism. These frames are passed through a Skeleton Pose Detection algorithm to map and isolate the user's physical joints. The spatial coordinates of these joints are mathematically normalized range to account for varying camera distances [15].

- Presentation Slide Pipeline (PPTX) Preprocessing:
  To transform the visual slideshow into structured semantic data for the Conversational AI, the system applies a strict four-step pipeline:

  a. *Extraction:* The system parses the .pptx internal structure to extract titles, body bullets, and hidden speaker notes. It converts tables into natural language sentences and applies Tesseract OCR to extract non-selectable text from embedded images [31].
  b. *Cleaning:* The raw text undergoes NFKC Unicode normalization. The system strips noise.
  c. *Assembly:* The cleaned text is chronologically ordered per slide (Title > Body > Image Captions > Speaker Notes). Repeated content is deduplicated, and all text is concatenated into a single, continuous "big story."
  d. *Chunking:* To optimize LLM context windows, the system utilizes NLTK to divide the "big story" into semantic passages of length 350 tokens. Each chunk is saved strictly in the CoQA (Conversational Question Answering) format [32].

3.2.2 Multimodal Emotion Recognition

- **Speech Emotion Recognition (SER):**
  From the preprocessed audio signal, the system initially extracts a comprehensive set of 190 high-dimensional acoustic features. This includes 80 Mel-Frequency Cepstral Coefficients (MFCCs)—covering lower-dimensional, delta, delta-delta, and standard deviations—to accurately capture the vocal tract shape and temporal speech dynamics. Additionally, the system extracts 36 Chroma features (STFT, CQT, CENS) to map tonal and harmonic content, 64 Log Mel Spectrogram (LMS) features to track intensity in the time-frequency domain, 6 Spectral Contrast features, 3 Root Mean Square Energy (RMSE) values, and 1 Zero Crossing Rate (ZCR) feature [1] to differentiate between voiced and unvoiced speech segments. To optimize computational efficiency and eliminate multicollinearity, Principal Component Analysis (PCA) is applied to this extracted set. The PCA effectively reduces the dimensionality of the feature space from 190 down to a concentrated 140 features that retain the most critical variance for emotion classification [30].

The 140 PCA-reduced features are fed into a sequence-based prediction framework utilizing a stacked Bidirectional Long Short-Term Memory (Bi-LSTM) network [30]. This architecture is specifically designed to deeply learn and fully capture the sequential and contextual dependencies within speech signals over time. The model is constructed using a deep sequential layering strategy:

a. *Temporal Feature Extraction Layers:* The network begins with three consecutive Bi-LSTM layers. Each Bi-LSTM layer utilizes 512 units to learn hierarchical and contextualized temporal features simultaneously from both forward and backward directions. To accelerate training and prevent overfitting, each Bi-LSTM layer is immediately followed by Batch Normalization, L2 regularization, and a Dropout layer with a rate of 0.3.
b. *Dense and Activation Layers*: Following the temporal extraction, the sequence data passes through two Dense layers, each containing 512 units. These layers utilize the Swish activation function to improve performance by mitigating the vanishing gradient problem, followed by an additional LeakyReLU activation to introduce necessary non-linearity.
c. *Output Classification:* The final output is generated by a Time Distributed layer that applies a Dense layer equivalent to the number of target classes (7 emotions) [30]. This layer utilizes a Softmax activation function to output the final emotion class probabilities for each time step, combined with a reshape layer to correctly represent the predicted emotional state.

- **Facial Emotion Recognition Module (FER):**
  Unlike speech processing which relies on handcrafted acoustic variables, the spatial hierarchies and emotional cues for this module are extracted directly from the preprocessed 64 times 64 three-channel facial tensors via deep convolutional layers. The network utilizes 3 times 3 convolution filters with a stride of 1 pixel across its blocks to efficiently capture nuanced spatial relationships and micro-expressions (such as subtle movements around the eyes or mouth). This specific kernel sizing allows the model to learn complex spatial patterns while sharing weights to maintain computational efficiency without requiring expensive processing overhead. Rectified Linear Unit (ReLU) activation functions are applied after each layer to introduce the necessary non-linearity for learning these complex visual features [10].

The extracted visual features are processed through the GiMeFive architecture, a custom 15-layer Convolutional Neural Network optimized for a precise balance between depth and computational performance, containing approximately 10.47 million parameters [10]. The model follows a strict sequential block architecture:

a. *Convolutional Blocks:* The network consists of 5 sequential Convolutional Blocks with progressively increasing filter depths to capture increasingly complex high-level features (64, 128, 256, 512, and 1024 filters, respectively).

b. *Regularization and Pooling:* Following each convolutional layer, Batch Normalization is applied to stabilize the learning process by ensuring forward-propagated signals maintain non-zero variances. This is immediately interleaved with 2x2 Max Pooling layers to systematically reduce the spatial dimensions of the input volume. To aggressively mitigate overfitting, a 20% dropout rate is applied after the max pooling layers in the first four blocks.

c. *Fully Connected (FC) Block*: After the fifth convolutional block, an Adaptive Average Pooling layer transitions the multi-dimensional feature maps into a 1D vector for the Fully Connected block. This block contains three linear layers: FC1 (2048 units), FC2 (1024 units), and a final FC3 output layer. A heavy 50% dropout layer is positioned between the first and second linear layers to further enforce generalization [10].

d. *Classification and Interpretability:* The final layer utilizes a Softmax function to exponentiate and normalize the logits into probabilities for multi-class emotion classification, optimized via Cross-Entropy loss. Crucially, to ensure transparency in the AI's decision-making, the model incorporates Gradient-weighted Class Activation Mapping (Grad-CAM). This mechanism leverages the 5th convolutional layer and a Global Average Pooling (GAP) layer to generate thermal heatmaps [10], visually highlighting the specific facial regions (e.g., warmer colors over uplifting mouth corners) that drove the model's final prediction.

- **Body Emotion Recognition Module:**
  From the preprocessed input skeleton structured as a tensor of 97 frames x 17 joints x (x, y) coordinates the module extracts three distinct physical hierarchies to capture movement dynamics. The pipeline isolates raw Joint positions, derives Bone vectors to map the structural relationships between those joints, and calculates Velocity to track the speed and trajectory of the user's physical movements over time [15].

The extracted features are processed through a Temporal Convolutional Neural Network (Temporal CNN) [15]. The architecture processes the three feature types (Joint positions, Bone vectors, and Velocity) through parallel, independent streams.

a. *Linear Embedding & Temporal Convolutions*: Each stream is first passed through a linear embedding layer to compress the features into 128 hidden units. The data then flows through 3 temporal scales to capture both micro-movements and overarching gestures: a fine scale (k=3, 3 blocks), a mid-scale (k=7, 2 blocks), and a coarse scale (k=15, 2 blocks).

b. *Concatenation & Attention*: The outputs of the three temporal streams are concatenated, resulting in a dense feature vector of 1152 dimensions per frame. An Attention Pooling layer applies softmax weights across the 97 frames to isolate the most emotionally significant temporal moments in the sequence [15].

c. Classification: The pooled data is passed through a Classifier head containing two linear layers (1152 >128 > 5). The final output classifies the physical emotion into one of 5 target classes: Anger, Fear, Happiness, Neutral, and Sadness.

3.2.3 Conversational AI & Feedback Engine

- **Question Generation Module (QG):**

To determine the appropriate context for questioning without relying on naive sequential reading, this module constructs a complex semantic graph directly from the provided text. The system extracts coreference clusters and named entities, connecting them to map logical keyword chains. By traversing this fully connected semantic graph, the system isolates interconnected sentences as "rationales" and extracts the specific target answer spans required to anchor the upcoming questions [20].

The formulation of the questions operates through a highly controlled, multi-stage generation pipeline:

a. *Question Type Classifier:* Before generation, a fine-tuned RoBERTa-large model acts as a binary classifier to explicitly dictate the question format. It processes the rationale, target answer, and conversational history to output a strict control signal—either <BOOLEAN> (for Yes/No queries) or <NORMAL> (for span-based queries).

b. *Conversational QG Pipeline:* The extracted features and control signals are passed into a fine-tuned T5-base generative model. The model concatenates the specific control signal, target answer, rationale, full context, and a shortened conversational history (capped at three previous turns) to generate a contextually coherent multi-turn question.

c. *Rewriting and Filtering (RF):* To enforce strict quality control, a secondary T5 model (trained for Conversational Question Answering) attempts to answer the newly generated questions. Questions are aggressively filtered out using heuristic rules if they yield incorrect answers, contain irrelevant tokens not found in the context, lack informative value, or repeat redundancies from previous turns [20].

d. *Cognitive Personas & Educational Framework:* To target varying levels of student comprehension, the module expands its generation capabilities by adopting distinct cognitive "personas" mapped directly to Bloom's revised taxonomy [22].

e. Model Selection: This pedagogical framework bypasses fine-tuning in favor of prompt engineering, utilizing state-of-the-art Large Language Models (LLMs) such as GPT-4 Turbo, GPT-3.5 Turbo, Llama-2-70B, Llama-3.1-405B, and Gemini Pro [20].

f. *Persona Generation:* The architecture forces the LLMs to simultaneously output six specific question variations for a single context, each fulfilling a specific cognitive persona: Remembering (factual recall), Understanding (explaining ideas), Applying (using information in new situations), Analyzing (drawing connections), Evaluating (justifying decisions), and Creating (producing original work) [24].

g. *Prompt Optimization:* To prevent cognitive misclassification—where a zero-shot model might confuse an "evaluating" persona with an "analyzing" persona—the module employs an 8-shot learning methodology. Injecting eight distinct context-question examples into the prompt completely eliminates redundancy and ensures the generated questions align precisely with the targeted educational objectives. Models like GPT-4 Turbo and Llama-3.1-405B specifically excel in this 8-shot setting, yielding perfectly distributed, taxonomy-aligned question sets.

- **Question Answer Validation (QAV):**

Zero-Shot Automatic Scoring Using DSPy: To evaluate user responses, the system employs an Automatic Short Answer Scoring with Feedback (ASAS-F) architecture. Specifically, it utilizes a strictly zero-shot approach (ASAS-F-Z), relying on the generalized reasoning capabilities of pre-trained LLMs to assess answers without requiring resource-intensive fine-tuning or labeled training datasets [24].

Automated Prompt Engineering & Output: To eliminate the instability of manual prompt manipulation, the zero-shot module is orchestrated via the DSPy framework. Instead of hardcoding prompts, the framework utilizes declarative signatures [24] that define the required input and output fields, allowing the system to automatically compile and optimize the instructions given to the LLM.

Given a context question, a reference answer, and the user's input, the ASAS-F-Z module outputs three distinct metrics:

a. A numeric score ranging from 0 to 1.

b. A categorical label classifies the response as correct, partially correct, or incorrect.

c. Detailed, elaborated textual feedback that explains the specific rationale behind the score, highlighting missing information or conceptual errors to guide user improvement.

- **Feedback Generation Module**

The Feedback Generation module transforms the outputs produced by the multimodal analysis and conversational evaluation modules into a unified, human-readable coaching report. The primary objective of this module is not only to identify weaknesses in presentation delivery, but also to provide constructive, psychologically supportive, and actionable guidance that helps users improve their communication skills over time.

The architecture follows a staged multimodal feedback pipeline composed of five conceptual components: Multimodal Inputs, Feature Extraction, a Reasoning LLM, a Coaching LLM, and the Final Feedback Output [26]. Raw audio, facial expressions, body posture signals, and question-answer evaluation metrics are first converted into structured behavioral indicators through feature extraction procedures. These indicators are then processed by a reasoning-oriented language model responsible for objective analytical interpretation, while a second coaching-oriented language model reformulates the analytical findings into supportive and motivational feedback suitable for the user [27].

Before higher-level analysis is performed, the audio stream is transcribed using the faster-whisper framework [24], which provides efficient speech-to-text conversion with precise timestamp extraction. The transcription output is segmented into sentence-level units to enable downstream linguistic and discourse analysis. In parallel, computer vision pipelines extract facial landmarks, posture alignment, emotional states, gaze behavior, and motion consistency metrics from video frames [36].

The reasoning stage computes multiple presentation-quality indicators, including emotional congruency, confidence-versus-anxiety indices, discourse coherence, speech pacing stability, filler-word density, and audience engagement consistency. Emotional congruency analysis evaluates whether facial, vocal, and body emotions remain behaviorally aligned throughout the presentation. For example, energetic speech combined with anxious facial expressions may indicate delivery inconsistency and reduced presentation confidence [35].

To improve interpretability and reduce hallucination during evaluation, the system adopts a Dual-Agent Feedback Architecture inspired by DIALEVAL [27]. The first agent performs strict analytical reasoning and extracts objective behavioral observations from the multimodal signals, while the second agent translates these findings into empathetic pedagogical feedback. This separation prevents emotional language generation from affecting the accuracy of the underlying analytical evaluation.

The final coaching stage follows the Sandwich Method feedback structure [38], where constructive criticism is placed between positive reinforcement and motivational encouragement. Additionally, each issue follows an Observation–Impact–Suggestion (OIS) structure. The system first identifies a measurable observation from the presentation data, explains its psychological or communicative impact on the audience, and finally provides a concrete actionable recommendation for improvement.

The generated report is formatted into a structured JSON schema containing praise summaries, constructive feedback entries, encouragement messages, and a fully formatted presentation report suitable for frontend visualization. This design enables seamless integration between the AI coaching backend and the user interface while ensuring consistency, modularity, and scalability of the feedback generation pipeline.

3.3 System Users

The Elevare platform is designed to bridge the communication gap across academic and professional environments. The primary user groups and their specific interactions with the system include:

- University Students: This group uses the system to prepare for classroom presentations, academic seminars, and project reports. They interact with the live video and audio feedback module to target surface-level and emotional delivery weaknesses, successfully reducing presentation-related anxiety and avoidance behaviors.

- Fresh Graduates and Job Applicants: These users operate the platform to practice pitching scenarios and mock job interviews. They utilize the automatic question generation and answer validation system to experience realistic, content-specific challenges based on their slides or resumes, transforming their technical competencies into clear, articulate delivery.

- Recruiters and Interviewers: Recruiters leverage the evaluation framework as an objective utility to conduct fair assessments of candidates. They review the fused multimodal reports detailing a candidate's body language, vocal variety, and facial expressions alongside standard performance metrics.

User Characteristics: to operate the project effectively, users must possess the following basic skills and experience:

- Baseline Presentation Awareness: Users should possess a basic understanding of the core topic or context material (such as their own presentation slides or text documents) that they intend to upload for customized question generation.

- Basic Technical Literacy: No specialized software development or technical engineering skills are required. Users only need the operational knowledge required to handle standard web interfaces, navigate an application GUI, and manage basic hardware inputs (such as a connected microphone and webcam stream).

- Core Soft Skill Objectives: The platform is designed for individuals tracking personal growth or requiring structural intervention for soft skill deficits, low presentation self-esteem, or public speaking anxiety.

3.4 System UI & User Manual

This section provides a comprehensive overview of the Elevare platform's User Interface. The interface is designed with a clean, modern aesthetic, ensuring that the user's focus remains on practicing and improving their public speaking skills through AI-generated analysis and feedback.

3.4.1 Registration Page

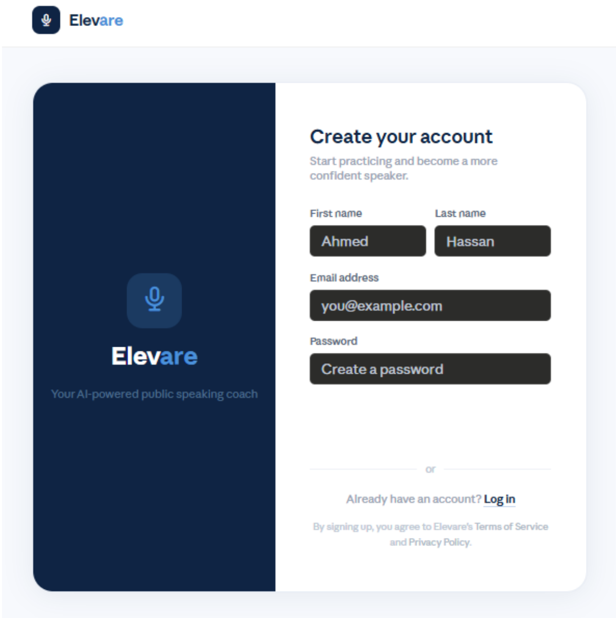

Figure 3.2 Registration Page.

The Registration Page serves as the entry point for new users joining the Elevare platform. It presents a straightforward account creation form that allows users to establish their credentials before accessing the system's full feature set. The layout reinforces the application's brand identity while guiding the user through the sign-up process with minimal friction.

- Branding Panel: The left side of the page features a dark navy panel displaying the Elevare logo accompanied by a microphone icon and the tagline "Your AI-powered public speaking coach," reinforcing the platform's purpose at first glance.

- Account Creation Form: The right panel presents a structured form collecting the user's first name, last name, email address, and password. Input fields are clearly labelled to prevent ambiguity.
- Login Redirect: A prompt reading "Already have an account? Log In" is included beneath the form, allowing returning users to navigate directly to the Login Page without confusion.
- Legal Compliance Notice: A footer note informs the user that by signing up, they agree to Elevare's Terms of Service and Privacy Policy, ensuring regulatory transparency.

3.4.2 Login Page

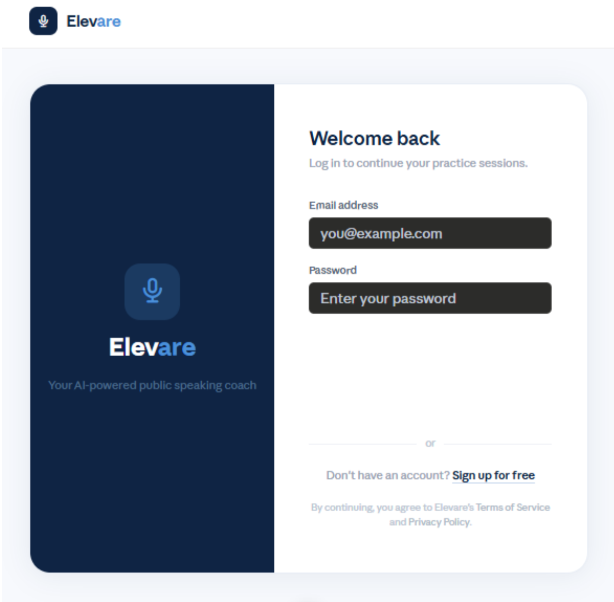

Figure 3.3 Login Page.

The Login Page provides authenticated access to the platform for returning users. It mirrors the visual language of the Registration Page, maintaining design consistency across the authentication flow. Upon successful login, the user is directed to the main dashboard.

- Branding Panel: Consistent with the Registration Page, the left panel displays the Elevare logo, microphone icon, and tagline, reinforcing the application's visual identity.
- Welcome Message: The heading "Welcome back" accompanied by the subtitle "Log in to continue your practice sessions" greets returning users in a personable manner.
- Authentication Form: The right panel contains fields for the user's email address and password, enabling credential verification against the system database.

- Registration Redirect: A prompt reading "Don't have an account? Sign up for free" is provided to guide new users toward the Registration Page, ensuring smooth bidirectional navigation within the authentication flow.

3.4.3 Home / Upload Page

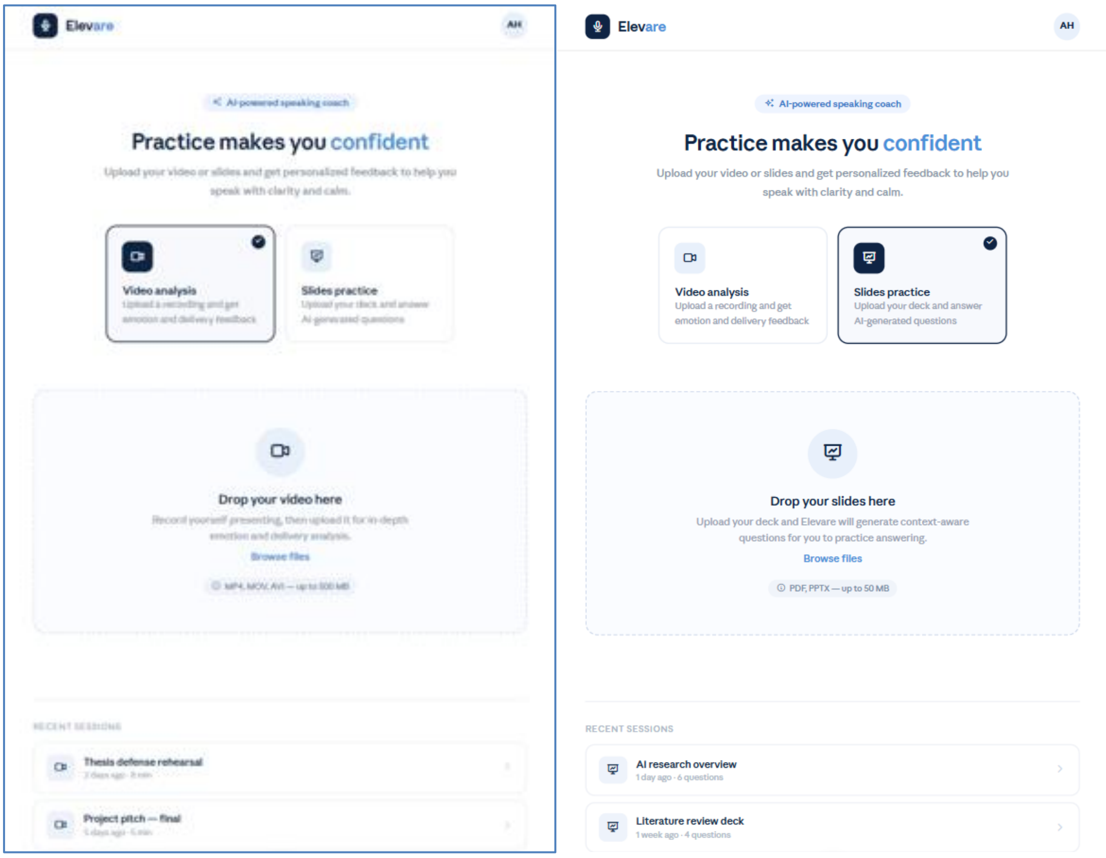

Figure 3.4 Home / Upload Page.

The Home Page serves as the central hub of the Elevare platform. Once authenticated, users are presented with two distinct practice modes: Video Analysis and Slides Practice. This page is designed to drive immediate engagement by presenting a clear call to action, and it supports drag-and-drop or file-browser-based content uploading. The page also retains a history of the user's past practice sessions for quick access.

- Navigation Bar: The top bar features the Elevare logo on the left and the authenticated user's avatar initials on the right, providing consistent identity confirmation throughout the session.
- Hero Section: A centered headline reading "Practice makes you confident" accompanied by a brief descriptive subtitle orients the user toward the platform's core value proposition.
- Mode Selection Cards: Two selectable cards allow the user to choose their practice mode. The "Video Analysis" card prompts the user to upload a recording to receive emotion and delivery feedback. The "Slides Practice" card allows the user to upload a presentation deck, after which the system

generates context-aware Q&A questions. The selected mode is highlighted with a dark background and a checkmark indicator.

- Upload Drop Zone: A dashed-border drop zone occupies the central area of the page. The zone dynamically adjusts its label and accepted file formats based on the selected mode. For Video Analysis, it accepts MP4, MOV, and AVI files up to 500 MB. For Slides Practice, it accepts PDF and PPTX files up to 50 MB. A "Browse files" hyperlink provides an alternative to drag-and-drop interaction.
- Recent Sessions: A section at the bottom of the page lists the user's previous practice sessions, displaying the session title, date, and duration or question count. This enables users to revisit prior work without navigating away from the home interface.

3.4.4 Video Analysis Results Page

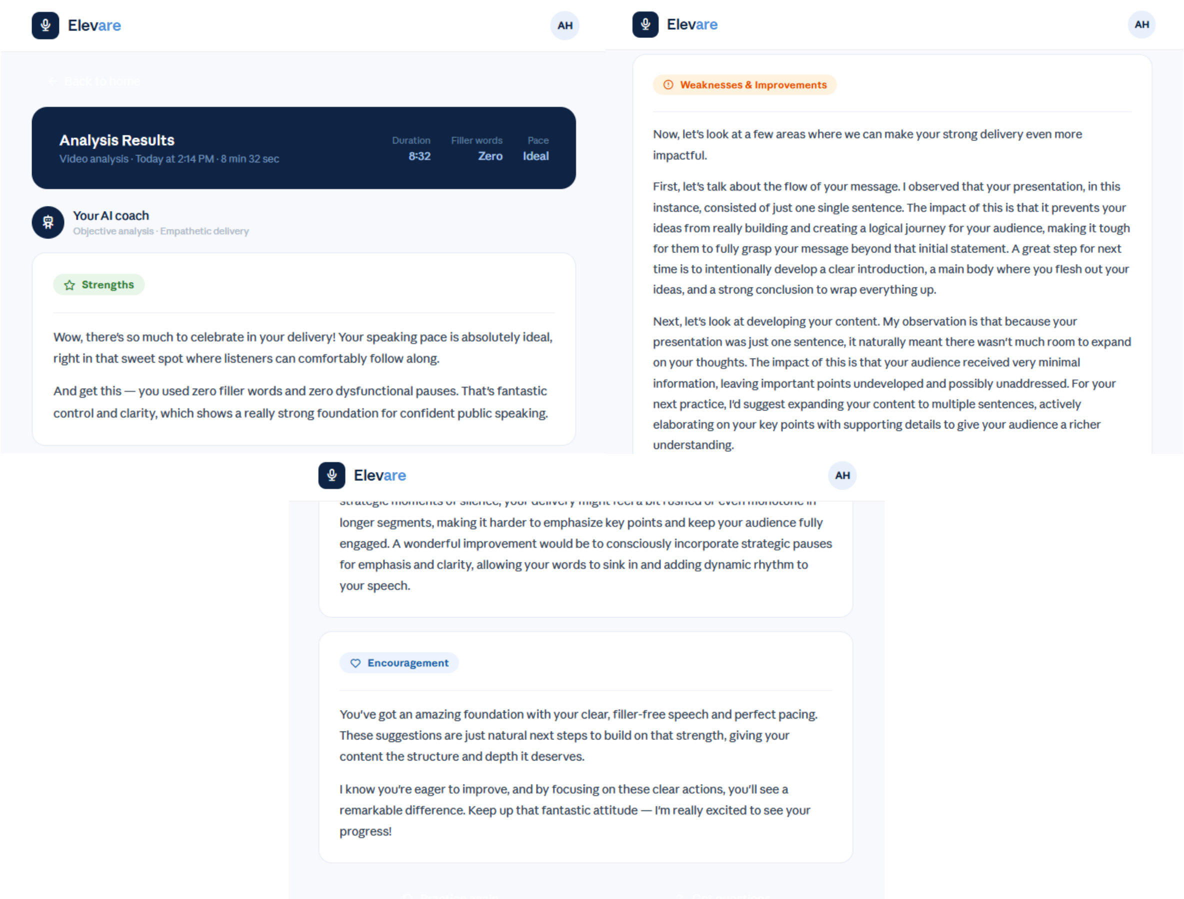

Figure 3.5 Video Analysis Results Page.

Upon the completion of video processing, the platform generates a detailed coaching report and presents it on the Video Analysis Results Page. This page focuses on readability and constructive formatting, ensuring that the AI-generated feedback is communicated in a clear, accessible, and encouraging manner. Key performance metrics are displayed prominently at the top of the page, followed by a structured breakdown of the user's delivery.

- Metrics Header Bar: A dark banner at the top of the results section displays session metadata alongside three key performance indicators: Duration, Filler Words count, and Pace rating. These metrics are colour-coded (e.g., green for ideal values) to provide immediate visual feedback.
- AI Coach Identity: A labelled avatar representing "Your AI coach" introduces the feedback section with the descriptors "Objective analysis" and "Empathetic delivery," establishing the tone of the coaching report.
- What Went Well: A green-tinted feedback panel highlights the strengths identified in the user's delivery, such as ideal pacing and absence of filler words, providing positive reinforcement.
- Areas to Improve: An amber-tinted panel presents targeted, actionable suggestions for improvement. Feedback is organized into distinct paragraphs addressing specific dimensions of the presentation, such as structural flow, content depth, and strategic use of pauses.

3.4.5 Q&A Session Page

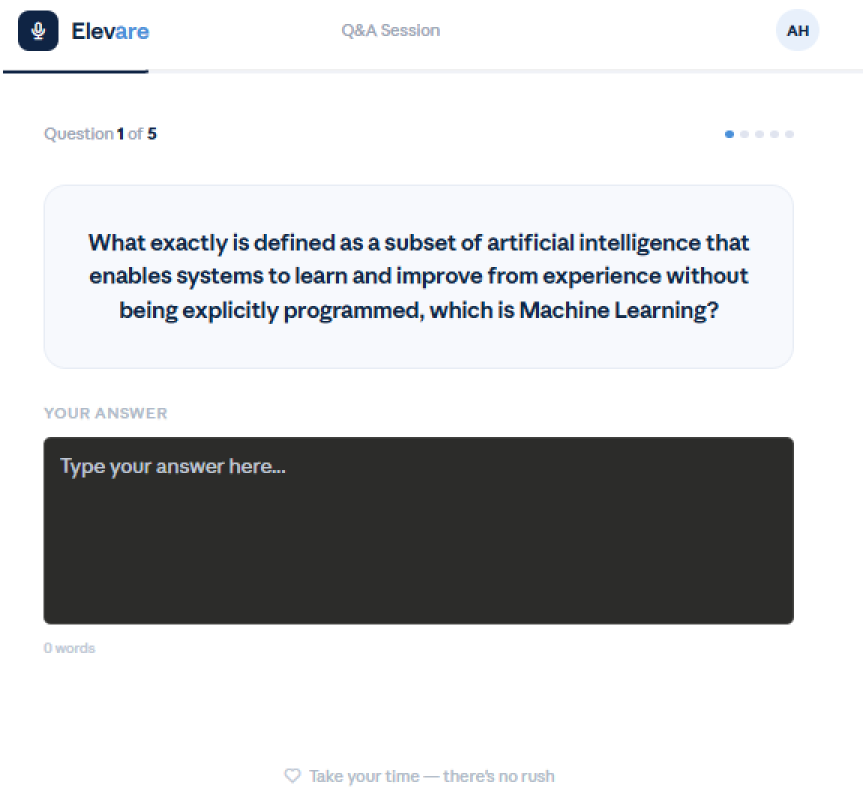

Figure 3.6 Q&A Session Page.

Following a slides upload, or as a standalone module, the Q&A Session Page presents the user with a series of AI-generated questions derived from the uploaded content. This view simulates an audience Q&A scenario to challenge the speaker's comprehension and on-the-spot articulation. The interface is intentionally minimalist to eliminate distractions and promote focused practice.

- Session Header: The navigation bar displays the session label "Q&A Session" in the center, alongside the Elevare logo and the user's avatar, maintaining interface consistency.
- Progress Indicator: The top of the content area displays the current question number relative to the total (e.g., "Question 1 of 5") alongside a dot-based progress tracker, giving the user a clear sense of advancement through the session.
- Question Display Card: The current question is rendered in a prominent, styled card with bold formatting, ensuring the user's attention is directed to the task at hand.
- Answer Input Area: A clearly labelled text area beneath the question card allows the user to type their response. A live word counter below the input field provides real-time feedback on response length.
- Encouragement Footer & Progression: A subtle line at the bottom reading "Take your time — there's no rush" reinforces a low-pressure environment. Upon completing their answer, the user proceeds to the next question, and upon answering all questions, is transitioned to the Q&A Grading Feedback Page.

3.4.6 Q&A Grading Feedback Page

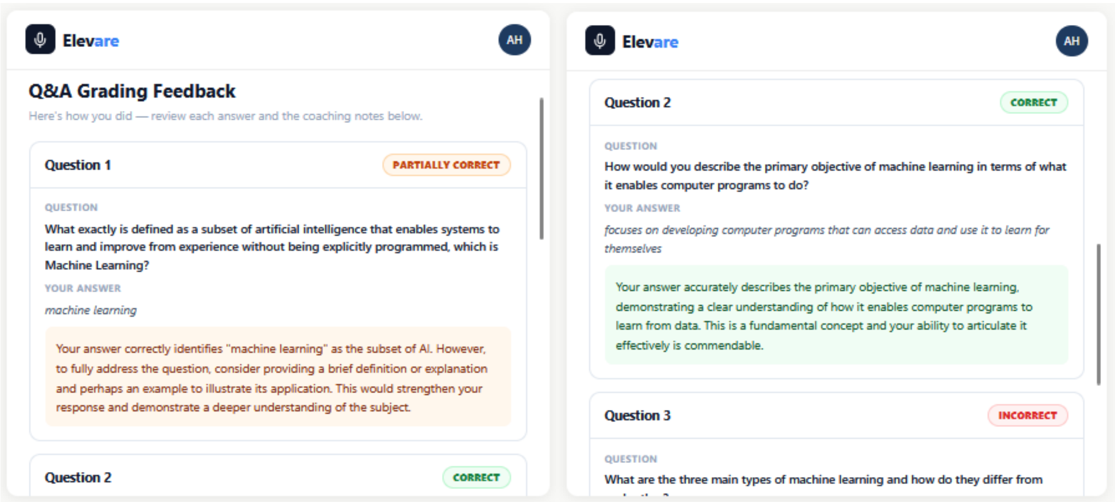

Figure 3.7 Q&A Grading Feedback Page.

The Q&A Grading Feedback Page constitutes the final stage of the Q&A practice workflow. It presents a consolidated, card-based assessment of the user's responses to each question in the session. The page is designed to deliver granular, constructive feedback while maintaining a clear visual hierarchy that allows the user to efficiently review their performance.

- Page Headline & Subtitle: A bold "Q&A Grading Feedback" title is positioned at the top of the page, accompanied by the subtitle "Here's how you did — review each answer and the coaching notes below," setting a constructive and encouraging tone.
- Card-Based Layout: Each question-and-answer pair is enclosed within a distinct, bordered card. This structure ensures clear visual separation between individual assessment items and improves the readability of the overall feedback report.
- Grading Badges: The upper-right corner of each card features a colour-coded status badge indicating the system's verdict on the user's response. Possible statuses include "CORRECT" (green), "PARTIALLY CORRECT" (amber/yellow), and "INCORRECT" (red), providing immediate and unambiguous performance signals.
- Detailed Breakdown Per Card: Within each card, the interface presents three labelled content sections: (1) Question — the original question posed during the session; (2) Your Answer — the verbatim text submitted by the user; and (3) Feedback — a detailed, AI-generated explanation of the grade assigned, including acknowledgment of correct reasoning and actionable suggestions for deeper elaboration where applicable.
- Color-Coded Feedback Panels: The feedback content within each card is rendered in a tinted background that mirrors the badge colour, reinforcing the grade visually and making it easy to distinguish correct, partial, and incorrect responses briefly.

Chapter 4
Implementation and Testing

Chapter 4: Implementation and Testing

4.1 Overview

This chapter presents the implementation details of the six core modules that constitute the Elevare system: Facial Emotion Recognition, Speech Emotion Recognition, Body Emotion Recognition, Question Generation, Question Answer Validation, and Feedback Generation. For each module, multiple modeling approaches were explored and are documented in order of experimentation. The datasets, preprocessing pipelines, feature extraction methodologies, and model architectures are described in detail, along with the empirical rationale behind key design decisions.

The technical breakdown begins with Facial Emotion Recognition (Section 4.2), which evaluates a ConvNext-based architecture augmented with a custom Detail Extraction Block against the GiMeFive convolutional network. This is followed by the Speech Emotion Recognition module (Section 4.3), built on a stacked Bidirectional LSTM network trained over acoustic features extracted from a combined multi-dataset corpus. Next, the Body Emotion Recognition module (Section 4.4) explores physical delivery through four progressively more sophisticated approaches: SVM, KNN, a Spatial-Temporal Transformer, and a Temporal Convolutional Network.

Shifting from behavioral tracking to material mastery, the Question Generation pipeline (Section 4.5) transforms presentation slides into structured conversational question-answer pairs, dynamically rephrasing them according to Bloom's Taxonomy cognitive levels. These generated queries are integrated with the Question Answer Validation module (Section 4.6), which utilizes a zero-shot LLM-based scoring approach to evaluate user responses against reference data. Concluding the system architecture, the Feedback Generation module (Section 4.7) fuses the multimodal emotion outputs and question-answer evaluation metrics into a single, coherent, and actionable performance report.

To provide empirical validation for these components, Section 4. summarizes the experimental testing results, detailing the trials conducted, configurations tested, and resulting performance metrics that justify the final design choices adopted in the deployed system.

4.2 Facial Emotion Recognition

4.2.1 FERPlus Dataset

The FERPlus dataset [37] is a large-scale facial expression recognition benchmark designed to provide high-quality emotion annotations for real-world images. It consists of 35,887 grayscale facial images, each originally collected from the FER2013 challenge [38], but later enhanced through crowd-sourced relabeling to improve the reliability of emotion categories. FERPlus includes eight emotion classes: Happiness, Surprise, Sadness, Anger, Disgust, Fear, Contempt, and Neutral. Unlike traditional single-label datasets, FERPlus captures the inherent ambiguity in human emotion perception by aggregating multiple annotators' votes for each image before assigning the final label. The dataset is organized into 28,709 training images, 3,589 validation images, and 3,589 test images, following the standardized split used across prior literature [37].

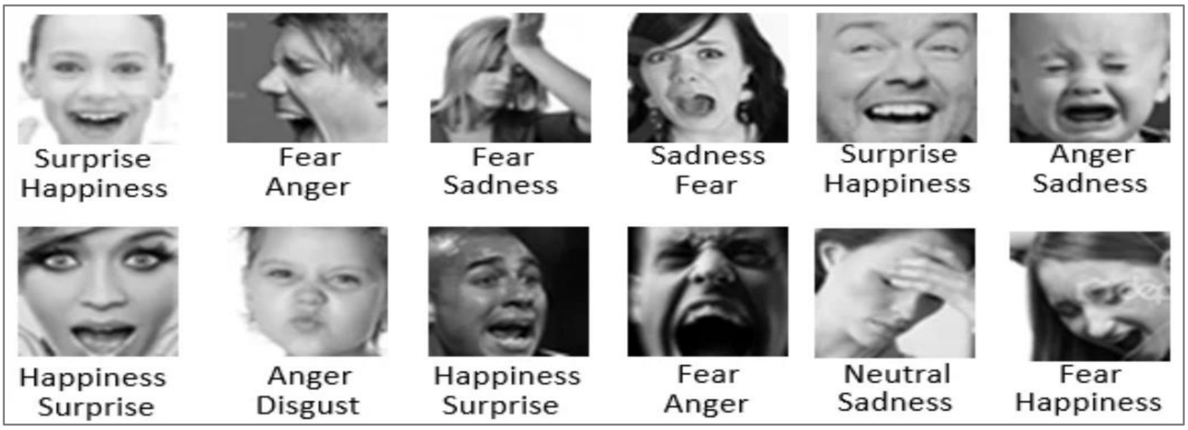

Figure 4. 1 Samples of the FERPlus dataset [37].

4.2.2 Preprocessing Pipeline used on FERPlus Dataset

- Preprocessing & Augmentation
  The FERPlus images were first standardized to ensure uniform input quality and compatibility with ImageNet-pretrained models. Each grayscale image was resized to 224×224, expanded to three channels, and normalized using ImageNet mean and standard deviation. During training, a comprehensive set of augmentation techniques was applied to improve robustness against the dataset's natural variability in pose, illumination, occlusion, and facial appearance. The augmentation strategy is summarized in Table 4.1.

Table 4. 1 Augmentation techniques applied during training.

Technique	Parameters	Purpose
Random horizontal flip	p = 0.5	Pose invariance
Rotation	±10–12°	Geometric robustness
Affine translation & scaling	~8% shift/zoom	Spatial variability
Random rescaling & translation	±20%	Expression boundary generalisation
Colour jitter	—	Lighting & sensor variation
Random erasing (cutout)	2–12% of image	Forces multi-region reliance

Additional augmentation techniques from prior FER literature — including heavy geometric distortion and region-level cropping — were evaluated during experimental phases but were ultimately reduced due to producing unrealistic distortions that negatively impacted model generalization [9,39-42].

- Class Imbalance Handling
  FERPlus exhibits significant class imbalance, with emotions such as Contempt, Disgust, and Fear being severely underrepresented relative to Happiness and Neutral. To address this, multiple complementary strategies were integrated into the training pipeline. Minority classes were oversampled using augmentation to increase their representation. Class-weighted loss was applied to penalize misclassification of underrepresented emotions more heavily. Additionally, LDAM (Label-Distribution-Aware Margin) loss combined with DRW (Deferred Re-Weighting) was adopted [43] to further improve decision boundary quality for rare classes. Training followed a two-stage warm-up strategy: the first stage used standard cross-entropy with the backbone frozen to stabilize gradient flow, followed by enabling LDAM+DRW and selectively unfreezing the 2nd and 3rd ConvNext stages to adapt the shared representation to the reweighted class distribution.

The feature extraction stage is built on a truncated ConvNext [33] backbone that captures low- and mid-level facial representations, combined with a custom Detail Extraction Block (DEB) designed to recover fine-grained expression cues. ConvNext is originally divided into four stages of progressively deeper semantic representations; however, for FERPlus — which contains subtle, small-scale variations across emotion classes — only the first two stages are retained. The higher semantic stages introduce unnecessary complexity and risk overfitting given the relatively small dataset size. This truncation results in a lightweight backbone focused on texture-level features such as micro-expressions, wrinkles, eyebrow curvature, eye shape, and lip motion [33].

To compensate for the reduced backbone depth, the DEB is placed after the truncated ConvNeXt to enhance sensitivity to subtle expression variations. It applies a sequence of depthwise separable convolutions [44] with progressively smaller kernels, followed by spatial dropout to prevent over-reliance on any single facial region, and a spatial pooling step to aggregate information across the entire feature map. A self-attention mechanism [33] is then applied to weight the most expressive facial regions — such as the eyes for fear, the mouth for happiness, and the eyebrows for anger — producing a compact yet discriminative feature representation that is passed to the classification head.

4.2.3 ConvNeXt + Detail Extraction Block Model

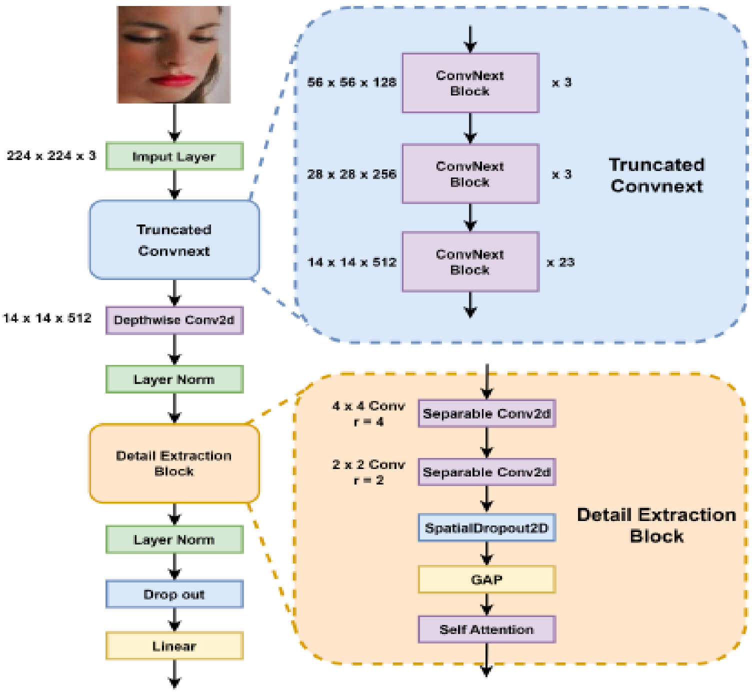

Figure 4. 2 ConvNext + Detail Extraction Block Architecture based on [9].

Figure 4.2 illustrates the end-to-end architecture of the proposed FER model. The pipeline begins with the truncated ConvNext backbone, which extracts low- and mid-level spatial features from the input image. These features are then passed through the Detail Extraction Block, which refines them using multi-scale depthwise convolutions, spatial dropout, and a self-attention mechanism, as described in the Feature Extraction section above. The refined feature maps are aggregated via global pooling and mapped to the eight FERPlus emotion categories through a fully connected classification head.

This design was chosen for its balance between accuracy, efficiency, and robustness under the specific constraints of FERPlus. Truncating the backbone reduces computational cost and overfitting risk while preserving the spatial detail most relevant to expression recognition. The DEB directly addresses the loss of fine-grained information that typically occurs in deeper convolutional stages, improving recognition of subtle and low-frequency emotions such as Contempt and Fear. The attention mechanism further improves reliability under occlusion, pose variation, and noisy annotations by directing the model toward the most discriminative facial regions [33].

The model was trained using the AdamW optimizer [45] with a weight decay of 1×10^{-4}. Mixed precision training (AMP) was enabled to improve computational efficiency, and gradient clipping with a maximum norm of 5.0 was applied to prevent exploding gradients. A differential learning rate strategy was adopted: the backbone was assigned a learning rate of 1×10^{-5}, while the Detail Extraction Block and classification head were assigned a higher learning rate of 1×10^{-4}. Training followed a two-stage schedule — a 5-epoch warm-up phase using standard Cross-Entropy loss with the backbone frozen, followed by LDAM+DRW training with the 2nd and 3rd ConvNeXt stages selectively unfrozen. Early stopping with a patience of 4 epochs was applied based on balanced accuracy to prevent overfitting.

4.2.4 RAF-DB Dataset

The RAF-DB (Real-world Affective Faces Database) dataset is a large-scale facial expression benchmark containing in-the-wild facial images collected under unconstrained real-world conditions. The official RAF-DB release totals 29,672 images across two subsets — a single-label subset (7 basic emotion classes) and a two-tab subset (12 compound emotion classes) — and is publicly available but requires a formal access request from a verified university-affiliated email, which we were unable to complete in time [46].

The Kaggle version was used instead [47], providing the same single-label subset without requiring this approval process: 12,271 training images and 3,068 test images across the same seven basic emotion categories (Anger, Disgust, Fear, Happiness, Neutral, Sadness, and Surprise).However, the Neutral class was excluded from our experiments, as its inclusion increased confusion and misclassification with other low-intensity expressions, with the model defaulting toward Neutral predictions in ambiguous cases. The resulting dataset used in this work therefore consists of six basic emotion categories: Happiness, Surprise, Sadness, Anger, Disgust, and Fear. This dataset exhibits significant class imbalance, with Happiness being the dominant class and Fear and Disgust being severely underrepresented. Minority classes were augmented to reach approximately 4,672 samples each to reduce class imbalance during training [48, 49].

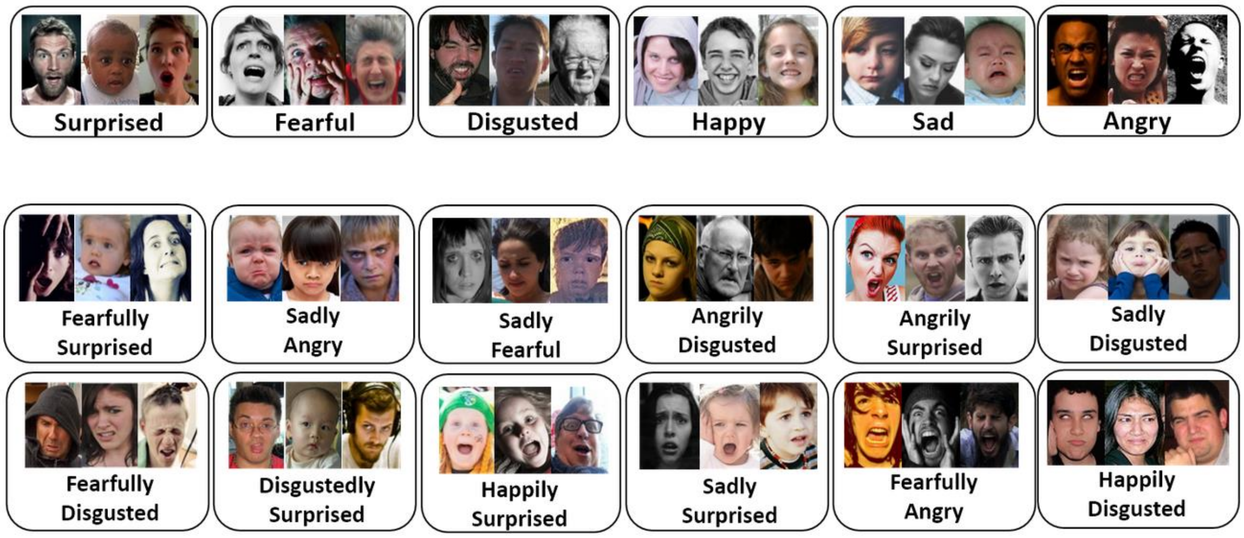

Figure 4. 3Sample images from the RAF-DB dataset across the six emotion categories [46].

4.2.5 Preprocessing Pipeline used on RAF-DB Dataset

- Preprocessing and Augmentation
  All input images were resized to 64×64 pixels and normalized using standard mean and standard deviation values consistent with the GiMeFive architecture's input requirements [10]. Given the significant class imbalance in the RAF-DB dataset, a class-aware augmentation strategy was adopted rather than applying uniform augmentation across all classes. Minority classes — specifically Anger, Disgust, and Fear — were heavily augmented to increase their representation, while majority classes received only light augmentation to preserve their well-represented feature distributions. The augmentation strategy is summarized in Table 4.2.

Table 4. 2 Class-aware augmentation strategy applied during training.

Technique	Majority classes	Minority classes	Purpose
Horizontal flip	✓ (light)	✓ (heavy)	Pose invariance
Rotation	±5°	±20°	Geometric robustness
Brightness jitter	±10%	±20%	Lighting variation
Random erasing	—	✓	Forces multi-region reliance
Mixup (α=0.2)	✓	✓	Inter-class boundary smoothing

Mixup augmentation interpolates between pairs of training images and their labels at α=0.2, encouraging the model to learn smoother decision boundaries between emotion classes [50]. This asymmetric class-aware strategy helped improve generalization across underrepresented emotion categories without degrading the model's performance on well-represented classes [51].

- Class Imbalance Handling
  The RAF-DB dataset presents a severe class imbalance challenge, with Happiness accounting for a disproportionately large share of the training data while Fear and Disgust represent only a small fraction. To address this, multiple complementary strategies were explored and integrated into the training pipeline.
  Augmentation-based oversampling was applied to minority classes to increase their sample diversity and reduce the risk of the model ignoring them entirely. Class-weighted loss was also evaluated, where higher penalties are assigned to misclassifications of underrepresented classes. However, applying strong class weights from the very first epoch was found to distort feature learning by preventing the model from forming stable representations before reweighting took effect.

  To overcome this, Deferred Re-Weighting (DRW) was adopted [43], which trains the model under standard Cross-Entropy loss for an initial period before introducing class weights in later epochs. This two-stage approach allows the model to first learn stable feature representations and then shift focus toward minority classes. Square-root-scaled class weights — proportional to $1/\sqrt{n_i}$ — were used to moderate the weight ratio between the rarest and most frequent classes, achieving meaningful minority class recall improvements without significantly degrading majority class performance [43].

4.2.6 GiMeFive Model

The model used in this work is the GiMeFive CNN, a custom 15-layer convolutional neural network proposed by Wang and Kawka [10] for facial emotion recognition. The architecture takes a 3-channel 64×64 image as input and produces probabilities over six emotion classes through five convolutional blocks followed by a fully connected classification head.

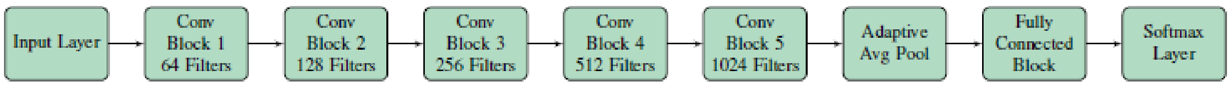

Figure 4. 4 GiMeFive model architecture based on [10].

Each convolutional block consists of a 3×3 convolution with padding=1, Batch Normalization, ReLU activation, 2×2 Max Pooling, and a Dropout layer at 20%. The five blocks progressively increase filter depth from 64 to 128, 256, 512, and 1,024 channels. Following the convolutional blocks, an Adaptive Average

Pooling layer aggregates spatial information, after which three fully connected layers map the learned representations to the output space: FC1 (2,048 units) → Dropout (50%) → FC2 (1,024 units) → FC3 (6 units). Final class probabilities are produced via a Softmax function. The architecture was selected for its strong balance between parameter efficiency and classification performance. With approximately 10.5 million parameters, GiMeFive is seven times smaller than VGG-16 [19] and roughly half the size of ResNet-34 [20], while achieving competitive accuracy on RAF-DB [10]. Squeeze-and-Excitation channel attention blocks were evaluated but excluded from the final configuration, as emotion recognition at 64×64 resolution was found to benefit more from spatial feature relationships than from channel-wise recalibration [10, 54]. The model was trained using the SGD optimizer with a learning rate of 0.001, momentum of 0.9, and weight decay of 1×10^{-4}, over 130 epochs with a batch size of 32 and a warm restarts [20] learning rate schedule with T_0=20 and T_mult=2.

4.3 Speech Emotion Recognition

4.3.1 Collected Dataset

The SER system was trained using a combined emotional speech dataset consisting of five publicly available speech emotion recognition datasets: RAVDESS, TESS, SAVEE, EmoDB, and CREMA-D. Combining multiple datasets improves emotional diversity, speaker variability, and model generalization. The final merged dataset contains speech samples from different genders, accents, and recording conditions.

Table 4. 3 Emotion class distribution across different datasets [30].

Dataset	Neutral	Happy	Sad	Angry	Fear	Disgust	Surprise	Total
Ravdess	96	192	192	192	192	192	192	1,248
Tess	400	400	400	400	400	400	400	2,800
Savee	120	60	60	60	60	60	60	480
EmoDB	79	71	62	127	69	46	0	454
CremaD	1,087	1,271	1,271	1,271	1,271	1,271	0	7,442
R+T+S	616	652	652	652	652	652	652	4,528
R+T+S+E+C	1,782	1,994	1,985	2,050	1,992	1,969	652	12,424

RAVDESS Dataset: The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) dataset contains 1,440 WAV audio recordings collected from 24 professional actors, including 12 males and 12 females. The dataset includes emotional speech samples representing calm, happy, sad, angry, fearful, surprise, and disgust expressions at both normal and strong emotional intensity levels. To maintain consistency with the combined dataset used in this work, the calm emotion category was excluded [56]

TESS Dataset:The Toronto Emotional Speech Set (TESS) dataset consists of 2,800 audio recordings captured from two female actors aged 26 and 64 years. The dataset includes seven emotional categories: happiness, anger, fear, disgust, pleasant surprise, sadness, and neutral emotion [57].

SAVEE Dataset:The Surrey Audio-Visual Expressed Emotion (SAVEE) dataset contains recordings from four male speakers aged between 27 and 31 years. The dataset includes emotional classes such as neutral, happy, sad, angry, disgust, fear, and surprise. However, the dataset suffers from class imbalance, as the neutral category contains nearly twice the number of samples compared to the other emotion classes [58].

EmoDB Dataset:The Berlin Emotional Speech Database (EmoDB) contains 535 German-language speech recordings categorized into seven emotional classes: anger, fear, sadness, happiness, disgust, boredom, and neutral. The recordings were sampled at 16 kHz with 16-bit resolution. Since the boredom class was not available in all combined datasets, it was excluded from this study [59].

CREMA-D Dataset:The Crowd-Sourced Emotional Multimodal Actors Dataset (CREMA-D) includes 7,442 audio recordings collected from 91 actors (48 males and 43 females) aged between 20 and 74 years. The dataset contains six emotion categories: happy, angry, neutral, sad, fear, and disgust, recorded at different emotional intensity levels including low, medium, high, and unspecified intensity [60].

4.3.2 Preprocessing Pipeline

The collected speech emotion datasets, including RAVDESS, TESS, SAVEE, EmoDB, and CREMA-D, underwent several preprocessing steps to improve data quality and ensure consistency before model training. Initially, silence removal was applied to reduce non-informative audio regions. Specifically, when the leading or trailing silence duration exceeded 200 ms, approximately 70% of the silence was removed to emphasize emotionally relevant speech content and reduce unnecessary noise. Furthermore, all audio clips were resampled to a unified sampling rate of 22,050 Hz to maintain consistency across the combined datasets. The waveform structures were also standardized after cleaning to ensure consistent signal characteristics during feature extraction. These preprocessing operations improved signal clarity, enhanced emotional feature representation, and reduced model distraction caused by silent frames and irregular audio segments [30].

To improve model robustness and reduce overfitting, multiple audio augmentation techniques were applied to the training dataset. Gaussian noise injection was performed using noise levels of 0.035, 0.025, and 0.015 to simulate real-world recording conditions and environmental noise. In addition, pitch shifting was applied in three stages using rate parameters of 0.70, 0.80, and 0.70, allowing variation in speaker vocal tone without altering the audio duration. Time stretching was also utilized with speed factors of 0.8, 0.9, and 0.7 to simulate different speaking rates while preserving pitch information. These augmentation techniques increased dataset diversity, helped address class imbalance issues, and improved the generalization capability of the SER model across different speakers and acoustic environments [50,61].

4.3.3 Speech Feature Extraction

Feature extraction is a fundamental step in SER because raw audio signals contain large amounts of redundant and unstructured information. To convert the speech signals into meaningful representations suitable for machine learning, several acoustic and spectral features were extracted from the audio recordings using the Librosa audio analysis library [62]. These extracted features capture temporal, spectral, tonal, and energy-related characteristics of speech that are strongly associated with human emotions. A total of 190 features were extracted from each audio sample to provide a rich and diverse representation of emotional speech patterns [30].

Table 4. 4 Extracted and selected feature List [30].

Feature	Description	Number of features
MFCCs	Temporal features of individual speech	20
MFCCs Delta	First derivative of MFCCs	20
MFCCs Delta2	Second derivative of MFCCs	20
MFCCstd	MFCCs standard deviation	20
Chroma STFT, CQT, CENS	Tonal and harmonic structure analysis	36
Log Mel Spectrogram	Captures both frequency and energy variations	64
Spectral Contrast	Measures the difference in peak and valley energy	6
Energy	Audio intensity of the signal	3
ZCR	Measures signal sign changes over zero	1

- Mel-Frequency Cepstral Coefficients (MFCCs)
  Mel-Frequency Cepstral Coefficients (MFCCs) are among the most widely used features in speech and emotion recognition systems. MFCCs represent the short-term power spectrum of speech signals while approximating the way humans perceive sound frequencies. They are

effective in capturing vocal tract characteristics and emotional variations in speech. In this work, 20 MFCC features were extracted from each audio sample, along with 20 delta MFCC features, 20 delta-delta MFCC features, and 20 MFCC standard deviation features. The delta and delta-delta coefficients represent the temporal changes in speech, enabling the model to capture speech dynamics and emotional transitions more effectively [63].

- Chroma Features
  Chroma features capture the tonal and harmonic characteristics of audio signals by mapping frequencies into 12 distinct pitch classes. These features are useful for identifying variations in pitch and harmonic structure that may correspond to emotional expressions in speech. Three types of Chroma features were utilized in this study: Chroma-STFT, Chroma-CQT, and Chroma-CENS. Each type captures tonal information differently, providing complementary representations of speech characteristics. A total of 36 Chroma-related features were extracted from each audio sample [64].

- Log-Mel Spectrogram (LMS)
  The Log-Mel Spectrogram (LMS) is a time-frequency representation of audio signals that combines spectral analysis with the Mel frequency scale, which closely resembles human auditory perception. LMS features provide detailed information about both the frequency distribution and energy variations within speech signals over time. These features are particularly effective for deep learning models because they preserve important emotional patterns in the speech signal. In this study, 64 Log-Mel Spectrogram features were extracted from each audio sample [65].

- Spectral Contrast
  Spectral Contrast measures the difference between spectral peaks and valleys within an audio signal. It helps identify variations between harmonic and non-harmonic components of speech and provides valuable information regarding speech quality and emotional intensity. Spectral Contrast is especially useful for distinguishing speech from background noise and identifying subtle emotional characteristics. A total of 6 Spectral Contrast features were extracted in this work [66].

- Root Mean Square Energy (RMSE)
  Root Mean Square Energy (RMSE) measures the average energy or amplitude of a speech signal over time. It reflects the intensity and loudness of speech, which are important indicators of emotional expression. Emotions such as anger or excitement often exhibit higher

energy levels, while sadness or calmness may produce lower energy signals. In this study, 3 RMS Energy-related features were extracted from each audio sample using the Librosa library [67].

- Zero Crossing Rate (ZCR)
  Zero Crossing Rate (ZCR) measures the number of times an audio waveform crosses the zero-amplitude axis within a given time frame. It is commonly used to differentiate between voiced and unvoiced speech segments and to analyze speech signal activity. ZCR is useful for identifying variations in speech patterns and capturing certain emotional characteristics present in audio signals. In this work, 1 ZCR feature was extracted from each audio sample [68].

4.3.4 Feature Selection and Data Splitting

After feature extraction, the extracted features were normalized using Z-score standardization (zero mean, unit variance). Principal Component Analysis (PCA) was then applied [69,70] to reduce feature dimensionality while preserving the most significant information from the extracted feature set. PCA helps minimize redundancy, reduce computational complexity, and improve model efficiency by selecting the most informative feature components. The extracted features were then normalized and divided into training and testing sets using an 80%-20% split ratio. Approximately 80% of the dataset was used for model training, while the remaining 20% was reserved for evaluation on unseen data. This splitting strategy ensures reliable performance assessment and helps evaluate the generalization capability of the proposed SER model.

4.3.5 BiLSTM Model

The model receives the PCA-reduced feature set as input, including representations derived from MFCCs, Chroma features, Log-Mel Spectrograms, Spectral Contrast, RMS Energy, and ZCR features. The core component of the architecture is the Bidirectional Long Short-Term Memory (BiLSTM) layer [71], which learns temporal dependencies in speech sequences from both forward and backward directions. Unlike traditional recurrent networks, BiLSTM can capture contextual emotional patterns occurring before and after a speech frame, making it highly effective for emotion recognition tasks where emotional cues evolve over time. A Dropout layer is included after the BiLSTM layer to reduce overfitting [72] and improve generalization by randomly deactivating neurons during training. The Dense layer then transforms the learned temporal representations into higher-level emotional embeddings that are more discriminative for classification.

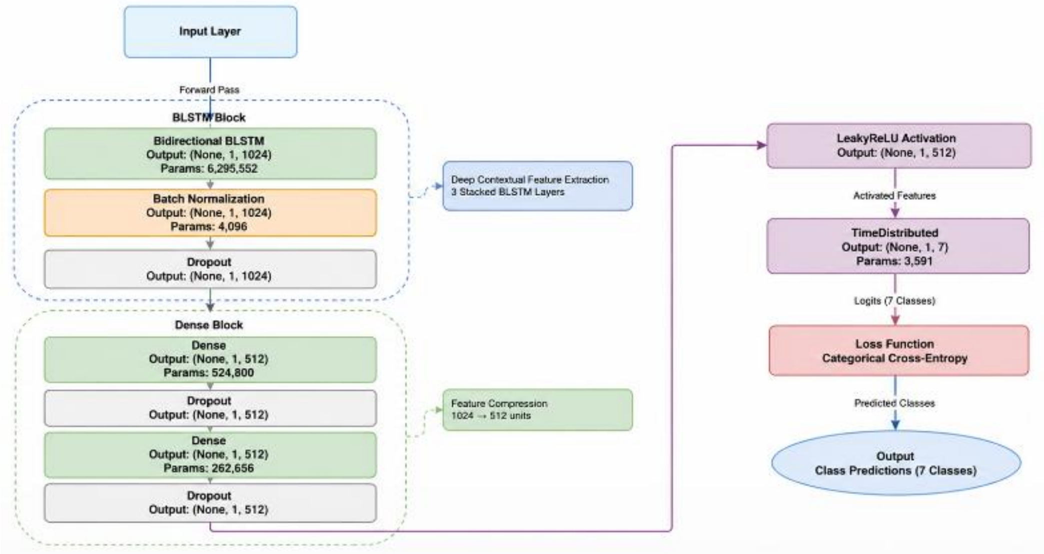

Figure 4. 5 BiLSTM Architecture adapted from [30].

Finally, the Softmax output layer produces a probability distribution across the target emotion classes, allowing the model to predict the most likely emotional state.The original architecture included a Deep Conditional Random Field (DeepCRF) layer to model sequential dependencies between output labels. CRF layers are generally beneficial in sequence labeling tasks because they consider relationships between neighboring predictions. However, during implementation, the input structure used a temporal dimension consisting of only one time step, which significantly limited the CRF layer's ability to learn meaningful transition dependencies between sequential outputs.

Additionally, the TensorFlow Addons implementation of the CRF layer is currently deprecated, which introduced additional implementation complexity and compatibility issues during development. Experimental evaluation showed that the inclusion of the CRF layer resulted in nearly identical classification performance compared to the BiLSTM-only architecture. This indicated that the BiLSTM layers alone were already sufficiently capturing the temporal and contextual information required for accurate emotion recognition.

Therefore, the CRF layer was removed to simplify the architecture and reduce computational overhead. This modification improved training efficiency, reduced model complexity, and enhanced maintainability without negatively affecting the overall classification performance or generalization capability of the system.The proposed architecture was selected because speech emotion recognition is inherently a sequential learning problem, where emotional information is distributed across time. BiLSTM networks are particularly effective in modeling long-range temporal dependencies and contextual relationships within speech signals. Additionally, the architecture balances strong feature learning capability with computational efficiency, making it suitable for handling large and diverse speech datasets. The model was trained using the Adam optimizer [53] with a learning rate of 1×10^{-4} and categorical cross-entropy as the loss function. A batch size of 256 was used over 500 epochs with early stopping [54] applied to prevent overfitting. The dataset was split into 80% training and 20% testing sets. Features were normalized prior to training, and PCA was applied for dimensionality reduction. The optimal PCA configuration was determined experimentally at 140 components.

4.4 Body Emotion Recognition

4.4.1 MEED Dataset

The multi-view emotional expressions dataset (MEED) dataset [75] is a multi-view skeleton-based corpus designed for emotion recognition from body expressions. It comprises 4,102 recordings captured across three camera angles (left, front, and right views) from 22 college student actors. Body pose is represented as 25 joints extracted via OpenPose. The dataset originally comprised seven emotion categories: Anger, Disgust, Fear, Happiness, Sadness, Surprise, and Neutral. In this work, two categories — Disgust and Surprise — were excluded as they were deemed inconsistent with the target recognition task, resulting in five retained classes: Anger, Fear, Happiness, Sadness, and Neutral.

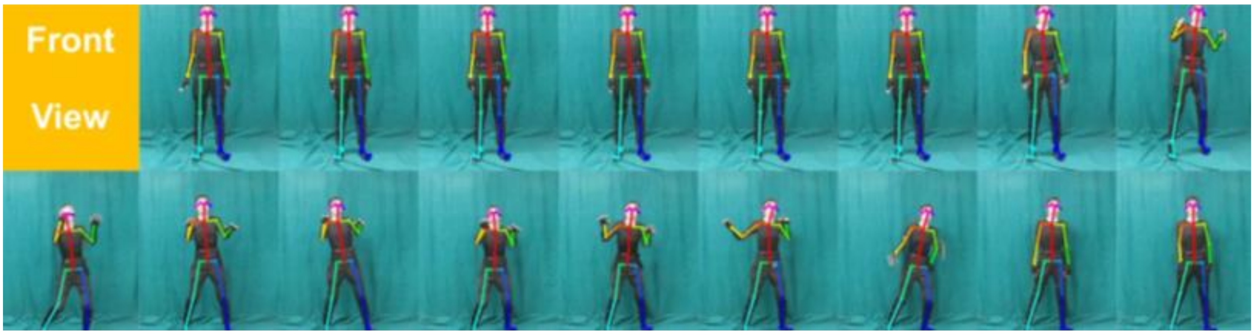

Figure 4. 6 Samples of front-view frames [75].

4.4.2 Shared Preprocessing Pipeline

All models in this module operate on OpenPose skeleton sequences extracted from the MEED dataset. The following preprocessing steps are common across all models.

- Confidence Masking & Interpolation
  Keypoints with a confidence score below 0.55 were treated as missing, consistent with values reported in prior work on keypoint filtering for pose-based recognition [76] and further validated through empirical trials on the training set. For models operating on sequential data, missing values were recovered through temporal interpolation across neighboring

frames. For frame-level models (SVM, KNN), low-confidence keypoints were masked as invalid and excluded from feature computation.

- Translation Normalization
  All joint coordinates were translated relative to the mid-hip joint to remove absolute position dependency. When the mid-hip was unavailable, the neck joint was used as a fallback anchor. Frames where neither anchor could be determined were discarded [77].

- Scale Normalization
  Coordinates were normalized by the shoulder width — defined as the Euclidean distance between the left and right shoulder joints — to ensure invariance to subject body size and camera distance. When shoulder joints were unavailable, a fallback scale was computed as the median distance of all valid keypoints from the center [78].

- Sequence Length Standardization
  All sequences were standardized to a fixed length of 97 frames, as the dataset provides videos of exactly 97 frames, except for a small subset containing slightly fewer frames [75]. Shorter sequences were padded by repeating the last frame to reach the standard length [79].

4.4.3 Body Emotion Classification Models

- Support Vector Machine (SVM)
  The shared preprocessing pipeline described in Section 4.4.2 was applied. To address class imbalance in the MEED dataset, Synthetic Minority Oversampling Technique (SMOTE) [80] was applied to the training set, generating synthetic samples for underrepresented emotion classes through feature-space interpolation.
  Following Vaijayanthi and Arunnehru [15], six pairwise Euclidean distances were computed from the normalized upper-body skeleton to capture arm extension and limb proximity to the torso: nose–right wrist, nose–left wrist, right wrist–right hip, left wrist–left hip, right elbow–right hip, and left elbow–left hip.

  SVM is a supervised learning algorithm that finds the optimal hyperplane maximizing the margin between classes in the feature space. For non-linearly separable data, the kernel trick implicitly maps input features into a higher-dimensional space where a linear decision boundary can be established. An RBF kernel was employed with regularization parameter $C = 1$ and kernel coefficient γ = scale. SVM was selected following

Vaijayanthi and Arunnehru [76], who applied SVM on the GEMEP dataset for body posture-based emotion recognition — a setting closely aligned with our MEED-based task.

- **K-Nearest Neighbors (KNN)**
K Nearest Neighbor (KNN) is considered among the simplest Supervised machine learning algorithms. The KNN algorithm is used for regression. However, it is used more often for classification. KNN is much faster than other algorithms because it doesn't learn or derive any discriminatory function during the training period. It keeps the training dataset and only learns from it in the prediction time. Thus, new data can be added easily without impacting the algorithm's accuracy. KNN is a non-parametric algorithm, as it doesn't assume any basic data.

KNN depends on the similarity between the new data and the training dataset that is previously stored. The new data is classified to the most similar class of the available classes, as shown in Figure 4.7. To determine the data category that is most like the new data.

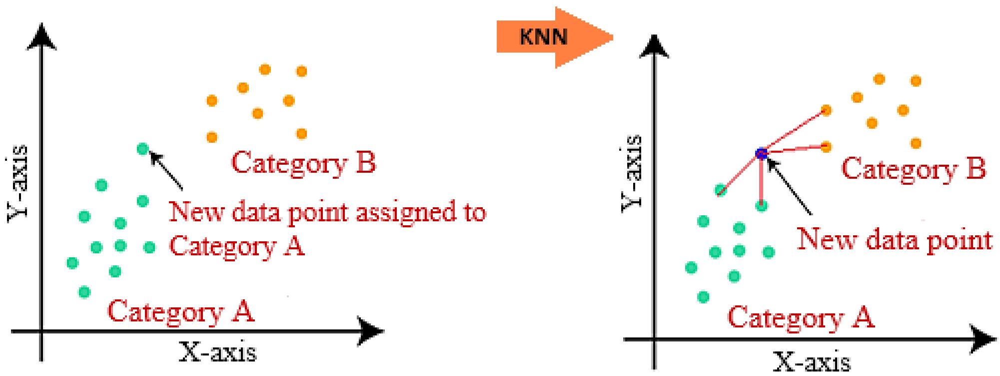

Figure 4. 7 K Nearest Neighbor Classification.

- **Spatial-Temporal Transformer (ST-TR)**
The shared preprocessing pipeline described in Section 4.4.2 was applied. Additionally, rotation normalization to remove orientation bias across recordings. A subject-discriminative joint subset was then selected, retaining only the 11 upper-body joints (indices 0–9 and 12) based on findings indicating that upper-body joints carry the most discriminative information for emotion recognition, reducing input dimensionality while preserving classification-relevant structure. Class imbalance was addressed through inverse-frequency class weights applied to the loss function during training.

Three feature channels were computed from the normalized joint subset per frame: raw joint positions (x, y coordinates), per-joint velocity (frame-to-frame displacement), and per-joint acceleration (second-order displacement).

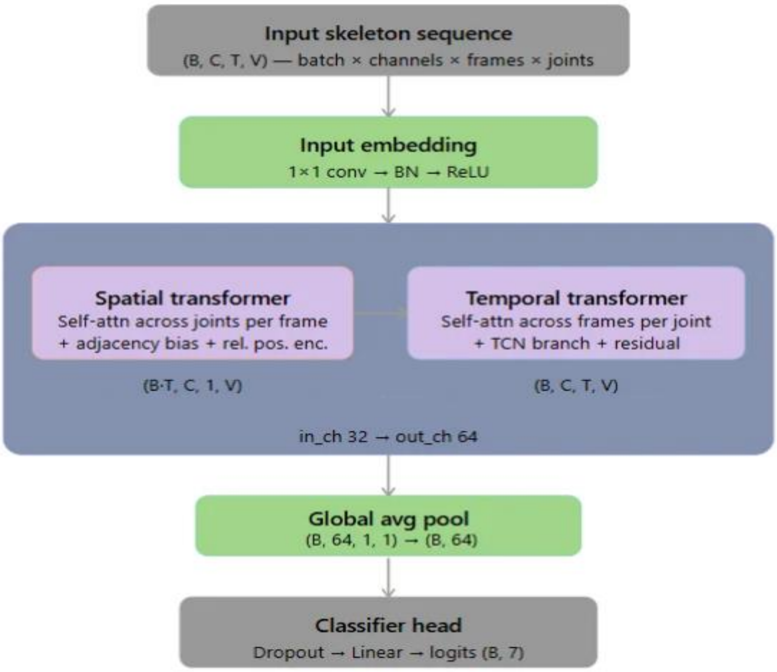

Figure 4. 8 ST-TR Architecture adapted from [81].

The ST-TR model [81] takes as input the skeleton sequence and projects it to a latent space via a 1×1 convolutional embedding followed by batch normalization and ReLU. The embedded features are processed through a dual-branch block operating in parallel: the spatial transformer applies self-attention across joints per frame, augmented with an adjacency bias and relative positional encoding to capture inter-joint relationships; the temporal transformer applies self-attention across frames per joint, combined with a TCN branch and residual connection to model motion dynamics. The output is aggregated via global average pooling and passed to a classifier head of dropout (rate = 0.5) followed by a linear projection to 7 class logits. The model was trained for 45 epochs with a learning rate of 5×10^{-4} on sequences of 97 frames.

ST-TR was selected over graph-based alternatives such as ST-GCN because its global attention mechanism captures correlations between distant body parts without being constrained by predefined skeletal connections — a property particularly relevant for emotion recognition, where expressive patterns may span non-adjacent joints. However, as the model was originally designed for action recognition, its architectural assumptions may not fully align with the subtler, holistic nature of emotion expression — motivating the exploration of architectures tailored more directly to sequential emotional body language in subsequent sections.

- **Multilayer Perceptron (MLP)**
  The shared preprocessing pipeline described in Section 4.4.2 was applied. Class imbalance was addressed through inverse-frequency class weights applied directly to the loss function during training, assigning higher penalty to underrepresented emotion classes.

  Following the findings of [82], which identified manual geometric features combined with fully-connected classifiers as an effective approach for skeleton-based emotion recognition on datasets structurally similar to MEED, five feature types were extracted per frame from the normalized skeleton sequence: pairwise Euclidean distances, joint angles, triangle areas formed by joint triplets, velocity, and acceleration. These features were concatenated per frame into a unified feature vector, which was then summarized across the temporal dimension using statistical aggregates (mean, standard deviation, median, energy) over fixed chunks, yielding a flat vector as the final model input.

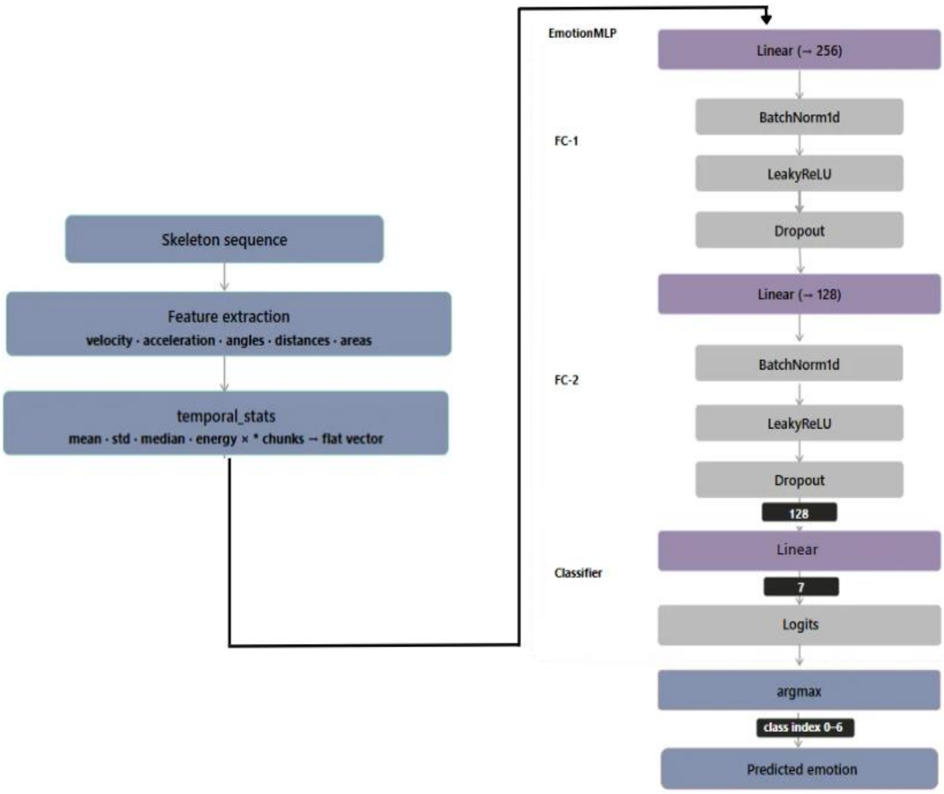

Figure 4. 9 MLP Architecture.

The proposed model is a fully-connected Multilayer Perceptron (MLP) consisting of two hidden layers of dimensions 512 and 256, each followed by a LeakyReLU activation and a dropout layer with rate 0.338. No batch normalization was applied. The output layer projects to 7 classes corresponding to the emotion categories in the MEED dataset, with the predicted class determined by argmax over the output logits. The network was trained using AdamW with a learning rate of 0.00313 and weight decay of 1.78×10^{-5}. A ReduceLROnPlateau scheduler halved the learning rate after 20 epochs without improvement, and early stopping was applied with a patience of 25 epochs. The loss function was weighted CrossEntropyLoss with inverse-frequency class weights.

The MLP was selected as a natural progression from the classical baselines, motivated by the survey's finding that manual features paired with fully-connected architectures yield strong performance on Emilya-like skeleton datasets. However, reducing the sequence to statistical aggregates discards the temporal order of body movement — motivating the use of architectures that explicitly model motion dynamics over time.

- **Temporal Convolutional Neural Network**
  In addition to the shared pipeline in Section 4.4.2, two model-specific steps were applied. First, the 25 OpenPose BODY_25 joints were remapped to the 17-joint H36M format [83] to standardize the skeleton representation; joints without a direct correspondence were approximated by averaging their OpenPose equivalents. Second, coordinates were normalized to the range [−1, 1] using the estimated image dimensions (720×576). To address class imbalance, data augmentation was applied during training, including horizontal flipping with symmetric joint swapping, Gaussian noise injection on coordinates, and random temporal speed perturbation.

  Three complementary feature streams were computed from the normalized skeleton sequence: joint positions (raw x, y coordinates), bone vectors (differences between connected joint pairs), and per-joint velocity (frame-to-frame displacement). Each stream captures a distinct aspect of motion static pose, structural relationships, and temporal dynamics respectively.

  The proposed model processes the three feature streams in parallel through independent branches. Each branch begins with a linear embedding layer projecting to 128 hidden dimensions, followed by multi-scale temporal convolutions at three kernel sizes: k = 3 (fine, ×3 blocks), k = 7 (mid, ×2 blocks), and k = 15 (coarse, ×2 blocks), capturing short, medium, and long-range temporal dependencies simultaneously. The outputs of all branches are concatenated per frame into a 1152-dimensional representation (384×3), which is then aggregated across the 97-frame sequence via attention pooling using softmax-weighted summation. The resulting vector is passed to a two-layer classifier head (Linear 1152→128, Linear 128→7) producing logits over the 5 emotion classes. The architecture and hyperparameters were determined through iterative experimentation.

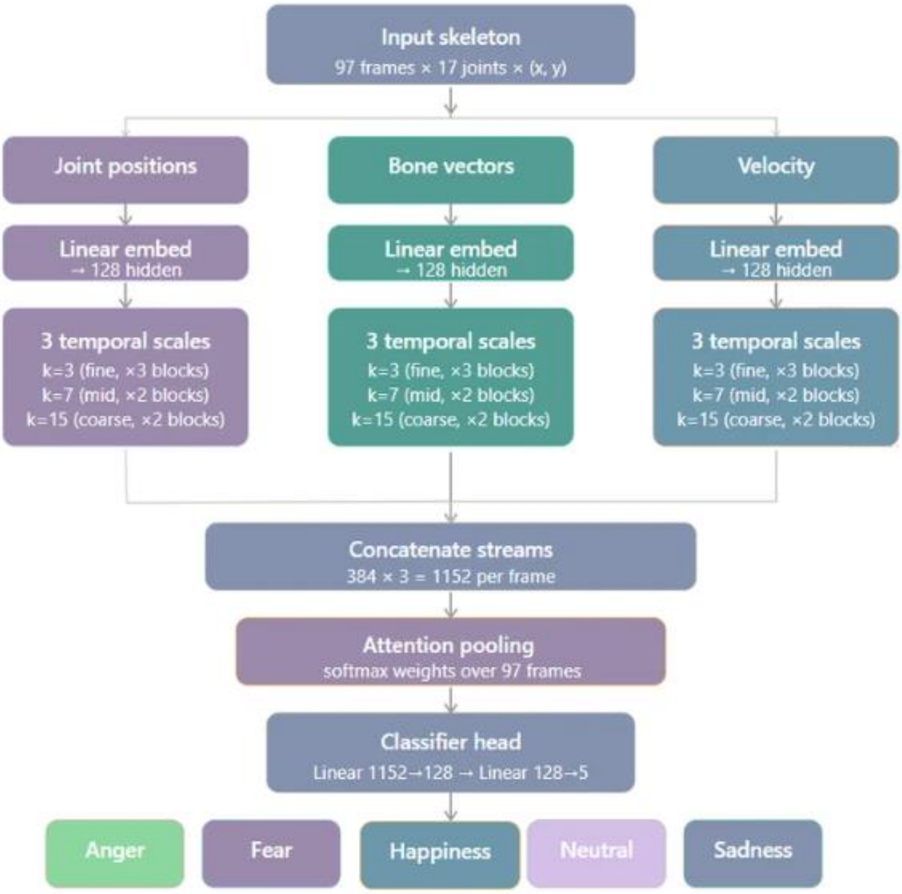

Figure 4. 10 Temporal CNN Architecture.

The Temporal CNN was motivated by the survey’s finding [82] that temporal convolutional approaches achieve strong performance on skeleton-based emotion recognition tasks structurally similar to MEED. Unlike the MLP, this model explicitly captures motion dynamics over time through multi-scale temporal convolutions, making it more suited to the sequential nature of emotional body expression.

4.5 Question Generation

4.5.1 CoQA Dataset

Conversational Question Answering Challenge (CoQA) [32] is a conversational question answering dataset designed to capture dependencies across multiple turns, containing over 127,000 questions from approximately 8,000 conversations spanning seven domains. Answers are provided as free-form text accompanied by evidence spans extracted from the passage. As shown in Figure 4.11, each sample consists of a passage followed by conversational question-answer pairs (Q_n, A_n), where questions may depend on previous turns, and each answer is supported by a rationale (R_n) extracted directly from the passage.

The Virginia governor's race, billed as the marquee battle of an otherwise anticlimactic 2013 election cycle, is shaping up to be a foregone conclusion. Democrat Terry McAuliffe, the longtime political fixer and moneyman, hasn't trailed in a poll since May. Barring a political miracle, Republican Ken Cuccinelli will be delivering a concession speech on Tuesday evening in Richmond. In recent ...

Q_1: What are the candidates **running** for?
A_1: Governor
R_1: The Virginia governor's race

Q_2: **Where**?
A_2: Virginia
R_2: The Virginia governor's race

Q_3: Who is the democratic candidate?
A_3: **Terry McAuliffe**
R_3: Democrat Terry McAuliffe

Q_4: Who is **his** opponent?
A_4: **Ken Cuccinelli**
R_4 Republican Ken Cuccinelli

Q_5: What party does **he** belong to?
A_5: Republican
R_5: Republican Ken Cuccinelli

Figure 4. 11 CoQA Sample [32].

4.5.2 Question Generation Preprocessing Pipeline

Prior to question generation, the input PowerPoint presentations are processed through a pipeline that extracts and structures textual content into CoQA-format contexts. CoQA was selected as the target format since the question generation models SQ and CQG are pretrained on CoQA, ensuring compatibility between the preprocessed input and the models' expected format.

- Shape Extraction
  Text is extracted from titles, body placeholders, and text boxes. Table content is converted into natural-language sentences by pairing column headers with their corresponding cell values. For image shapes, OCR is applied using Tesseract [84] to extract embedded text. Speaker notes are appended as additional context.

- Text Cleaning:
  Extracted text undergoes Unicode normalization and control character removal. URLs and lone slide numbers are discarded. Logical symbols are mapped to their textual equivalents, and bullet points are split into individual sentences.

- Story Assembly

Content from each slide is ordered as: title, body, image captions, and speaker notes. Duplicate sentences are removed and all slides are concatenated into a single continuous passage.

- Passage Chunking

  The passage is split into chunks of at most 350 tokens using NLTK sentence tokenization [85], preserving sentence boundaries. The 350-token limit was chosen to match the average context length in the CoQA dataset [32], ensuring the chunked input remains consistent with the distribution the models were trained on. Each chunk is saved in CoQA format ready for input to the question generation stage

4.5.3 Question Generation Model

The question generation module consists of two sequential stages: conversational question generation using the SG-CQG framework, followed by persona-based rephrasing guided by Bloom's Taxonomy.

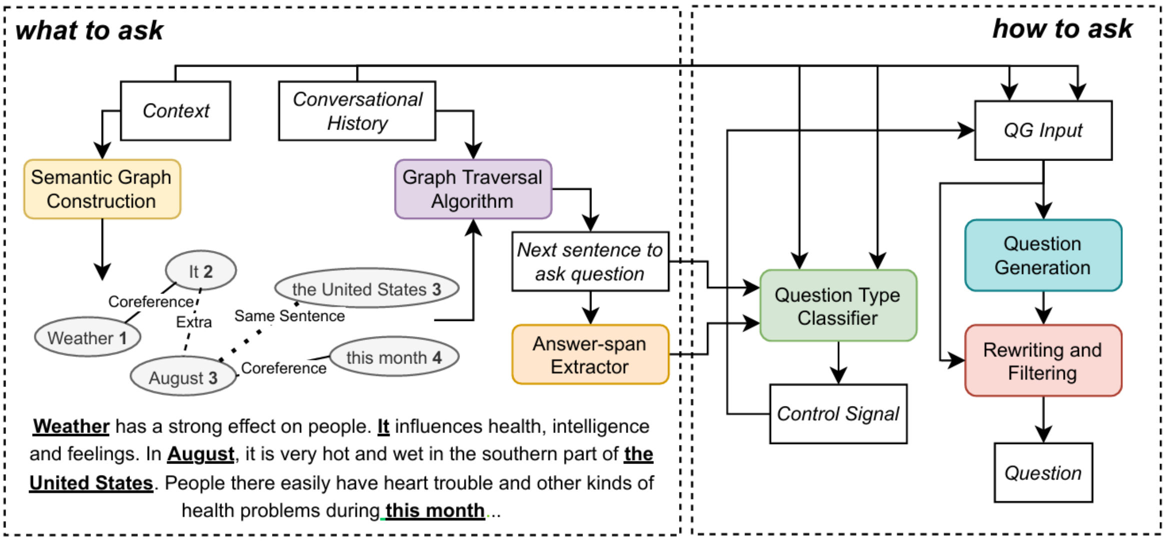

Figure 4. 12 G-CQG model architecture [86].

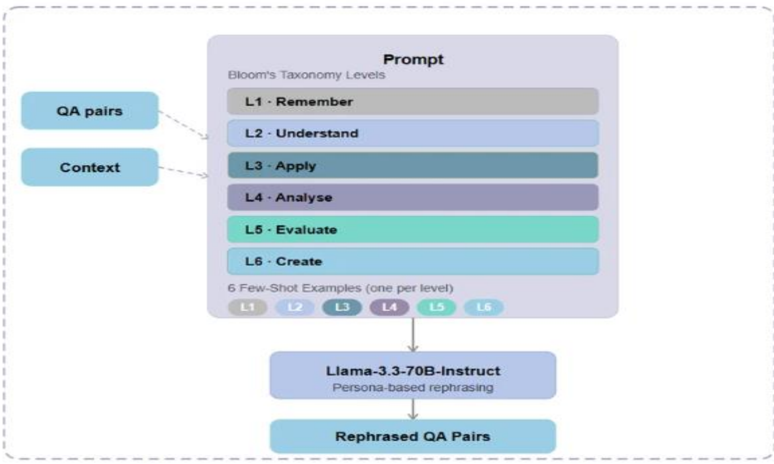

Figure 4. 13 Bloom's Taxonomy rephrasing module.

- Stage 1: Conversational Question Generation (SG-CQG)
  Following the SG-CQG architecture proposed by Westera et al. [86], the question generation stage operates through two modules.

  What-to-Ask: Given an input context, the model constructs a Semantic Graph by identifying named entities and resolving coreferences, linking pronoun references back to their respective entities. This graph encodes the relational structure of the passage. A graph traversal algorithm then operates over this structure, incorporating the conversational history to select the most appropriate rationale — the sentence or concept that grounds the next question — ensuring that questions follow the logical flow of the document.

  How-to-Ask: Once a rationale is selected, the how-to-ask module determines the form of the question. A question type classifier first produces an explicit control signal specifying the target question category — for instance, distinguishing between boolean yes/no questions and open-ended wh- span questions. This control signal, together with the selected rationale and the QG input, is passed to a fine-tuned T5 transformer [5] that synthesizes the final natural language question. The T5 generator ensures that the output is both grammatically well-formed and semantically grounded in the source context.

- Stage 2: Persona-Based Rephrasing via Bloom's Taxonomy
  While Stage 1 produces factually grounded questions, real audiences are not uniform different individuals approach the same material with different backgrounds, intentions, and cognitive goals. To capture this diversity, the QA pairs produced by SG-CQG are passed to a rephrasing stage that transforms them into persona-specific questions.

  Bloom's Taxonomy [88] is a hierarchical framework for classifying educational objectives according to cognitive complexity. As shown in Table 4.5, it defines six levels in ascending order: Remember (L1), retrieving specific facts from memory; Understand (L2), interpreting and explaining meaning in one's own words; Apply (L3), using knowledge in a new practical situation; Analyze (L4), breaking down material into its components and examining implications; Evaluate (L5), making critical judgements against established criteria; and Create (L6), generating novel ideas, designs, or hypotheses. Each level builds upon the previous, progressing from pure recall toward higher-order thinking.

Table 4. 5 Bloom's Taxonomy [88].

Bloom's level	Description
Remember	Retrieve relevant knowledge from long-term memory.
Understand	Construct meaning from instructional messages, including oral, written, and graphic communication.
Apply	Carry out or use a procedure in a given situation.
Analyse	Break material into foundational parts and determine how parts relate to one another and the overall structure or purpose.
Evaluate	Make judgements based on criteria and standards.
Create	Put elements together to form a coherent whole; reorganise into a new pattern or structure.

Bloom's Taxonomy was chosen because its six cognitive levels align naturally with the audience of personas defined in this work — from a beginner who recalls basic facts (Remember) to an innovator who synthesizes new ideas (Create). Alternative frameworks, such as PersonaGym [89], define personas around demographic and lifestyle attributes — for example, "A 36-year-old environmental lawyer from Australia, fighting against illegal deforestation and protecting indigenous lands" — which are not relevant to the educational and cognitive goals of this system.

Each persona is mapped to one of the six Bloom levels, and a structured prompt is constructed specifying the characteristics and objectives of each level, along with six few-shot examples — one per level — to guide the model's output style. This prompt, together with the original QA pairs and the source context, is submitted to LLaMA 3.3 70B Instruct, which rephrases each question to reflect the intent of the corresponding persona.

The SG-CQG module was selected due to its explicit question type classifier, which enforces control over the output question category, ensuring diversity and avoiding the disproportionate number of boolean yes/no questions common in implicit generation models. Additionally, by representing the input context as a Semantic Graph rather than a flat sequence, the model captures relational structure and coreference links between entities, producing questions that are logically grounded in the document's information flow. LLaMA 3.3 70B Instruct was selected for the rephrasing stage as the most capable openly available instruction-tuned model, making it well-suited for reliably following persona-specific prompts across six cognitive levels.

4.6 Question Answer Validation

4.6.1 SAF Dataset

The Short Answer Feedback (SAF) dataset [90] was used for evaluation in this module. The dataset consists of 29 short-answer questions drawn from the domain of communication networks, with a total of 9,732 student answers. Each entry includes a reference answer, a numeric score, a label (Correct, Partially Correct, or Incorrect), and human-written elaborated feedback. The SAF dataset is a publicly available benchmark that, notably, pairs scoring with content-focused feedback, making it particularly suited for evaluating both scoring accuracy and feedback quality simultaneously [31].Due to the domain mismatch between the SAF dataset's communication networks content and our project's focus on presentation skills assessment, the dataset was used exclusively for testing and not for any form of training or fine-tuning.

4.6.2 Approach Selection

four techniques were evaluated for automatic short answer scoring with feedback: zero-shot Large Language Model (LLM) prompting (ASAS-F-Z), automatic few-shot prompt optimization (ASAS-F-Opt), Retrieval-Augmented Generation (RAG)-based few-shot scoring (ASAS-F-RAG), and a similarity-based majority-vote baseline. ASAS-F-Opt uses automatic Bayesian prompt optimization with labeled examples [91], ASAS-F-RAG retrieves similar labeled examples at inference time using ColBERT [92], and the majority-vote baseline assigns scores based on similarity to training examples. Since the paper's results are tied to the communication-networks domain and our use case involves open-ended, unpredictable presentation questions with no available labeled examples, we conducted independent research to determine the most suitable approach for our context [93, 94, 95].

The comparison between zero-shot and few-shot RAG approaches revealed that zero-shot generalizes best to new and unseen questions [31,93], making it the most appropriate choice for our setting where question structure is variable and no retrieval database of labeled examples is available. Few-shot RAG, while producing more consistent and human-aligned outputs, requires a maintained database of labeled examples and a retrieval pipeline, both of which are unavailable in our deployment context [31,96]. Zero-shot is also more appropriate here because presentation questions are open-ended and highly variable, meaning retrieved examples from a fixed database would rarely be semantically relevant enough to guide the model correctly. Accordingly, the zero-shot ASAS-F-Z approach was adopted as the final system.

4.6.3 Zero-Shot (ASAS-F-Z) Model

The module is defined through a DSPy Signature named Student Answer Scoring, which formally specifies the input fields question, student answer, and reference answer — and the expected output fields a label, a numeric score, and an explanation. This signature is passed to a Typed Predictor module, which enforces strict type constraints on the model output, ensuring the response adheres to the predefined structure and invokes the model in a zero-shot manner without any few-shot examples or fine-tuning [97].

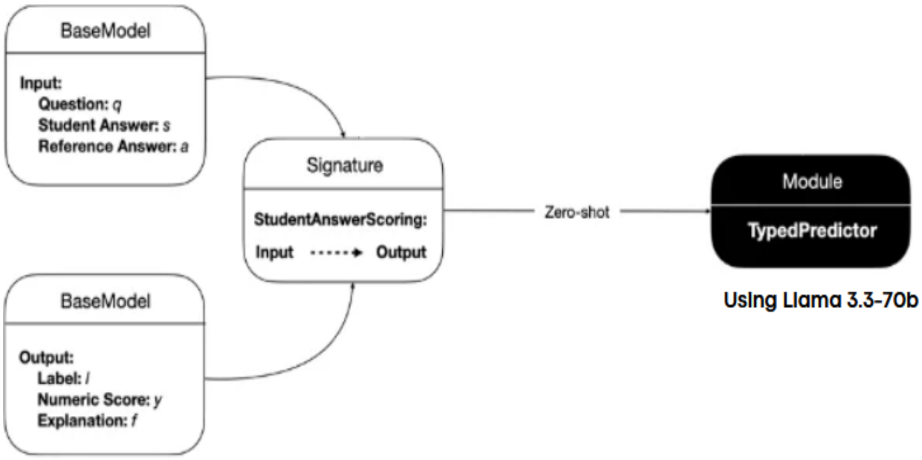

Figure 4. 14 High-level architecture of the Zero-Shot ASAS-F-Z module implemented with DSPy and Llama 3.3-70B [31,97].

Figure 4.13 illustrates the high-level design of the zero-shot scoring module. The system is built using the DSPy (Declarative Self-improving Python) framework [97], a tool that automates prompt generation and refinement by allowing developers to define input/output signatures rather than hand-crafting prompts. This eliminates the need for complex prompt engineering and makes the system more modular and adaptable to different question types. The backbone language model used is Llama 3.3-70b [98], accessed via the Groq API [99].

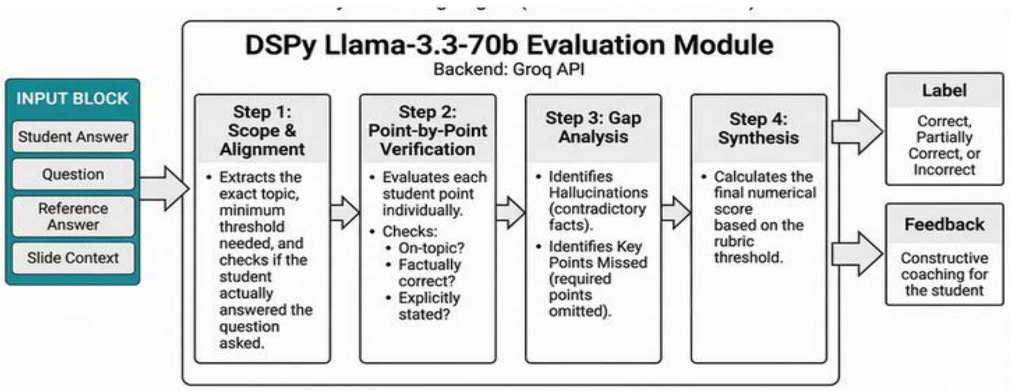

Figure 4. 15 Internal reasoning pipeline of the DSPy Llama 3.3-70B evaluation module [31,97].

Figure 4.14 illustrates the internal reasoning pipeline of the evaluation module, which consists of four sequential steps. In the first step, Scope and Alignment, the model extracts the exact topic of the question, determines the minimum score threshold, and verifies that the student answer actually addresses the question asked. In the second step, Point-by-Point Verification, each student point is evaluated individually for topical relevance, factual correctness, and whether it was explicitly stated. The third step, Gap Analysis, identifies any hallucinations or contradictory facts present in the student answer, as well as key points that were required but omitted. Finally, in the fourth step, Synthesis, the model calculates a final numeric score based on the rubric threshold and produces two outputs: a Label (Correct, Partially Correct, or Incorrect) and constructive Feedback for the student. This multi-step reasoning structure was adopted to reduce hallucinations, improve scoring consistency, and ensure each student point is evaluated independently before a final score is synthesized [31,97].

Llama 3.3-70b was selected as the backbone model based on both performance and practical constraints. While Mistral-7b achieved slightly higher accuracy and F1 scores in the [31]'s zero-shot evaluation, Llama 3.3-70b achieved the best Root Mean Square Error (RMSE) across datasets, indicating more stable and reliable numeric scoring on unseen questions. RMSE measures the deviation between predicted and actual numeric scores, making it a critical metric for a scoring system where numeric precision directly impacts feedback quality. Additionally, Llama 3.3-70b was fully available at no cost within our deployment environment, whereas higher-performing Mistral variants required paid API access.

4.7 Feedback Generation

The feedback pipeline consists of five conceptual components: Multimodal Inputs, Feature Extraction, a Reasoning LLM, a Coaching LLM, and the Final Feedback output. Raw audio and video are first transformed into structured, machine-readable metrics during the Feature Extraction stage. These metrics are then passed to the Reasoning LLM, which performs objective, data-driven analysis free from emotional framing, and finally to the Coaching LLM, which translates the analytical output into encouraging, constructive feedback for the user [100].

Figure 4. 16 Feedback pipeline.

4.7.1 Multimodal Feature Extraction

Before advanced features are calculated, raw audio is transcribed using a dedicated processing block built on faster-whisper [37], a reimplementation of OpenAI's Whisper model [101] using the CTranslate2 inference engine. The WhisperModel is instantiated with configurable model sizes and precision types, such as int8 quantization, to transcribe audio into precise word-level timestamps, text segments, language probabilities, and exact duration metrics. faster-whisper was selected over the original Whisper implementation due to its significantly faster inference speed and lower memory footprint while preserving comparable transcription accuracy. Following transcription, sentence boundaries are extracted using a regex pattern that splits continuous text based on trailing punctuation, enabling sentence-level analysis in later stages.

The features extracted in this stage were selected based on findings in communication science research linking specific verbal and vocal behaviors to perceived speaker quality. Speech disfluencies, including filler words and unplanned pauses, have been shown to negatively affect perceived speaking effectiveness, particularly at higher rates [109]. Speaking rate (WPM) similarly affects how thoroughly an audience processes and is persuaded by a message, with both very slow and very fast pacing shown to reduce a listener's ability to engage with the content [110]. Vocal acoustic properties — including pitch, intonation, and loudness — are well-documented paralinguistic cues that listeners use to judge a speaker's confidence and credibility [111]. Eye contact has likewise been linked to increased perceived credibility and persuasiveness in public speaking contexts [112]. These features were therefore selected as objective, measurable proxies for communication qualities established in the literature as meaningfully affecting how a presentation is received. From this transcription, a comprehensive set of features is extracted, broadly grouped into four categories:

- Speech Features: Words Per Minute (WPM), filler word detection, pause analysis distinguishing functional from dysfunctional pauses, and stutter recovery detection.
- Emotion Analysis: Speech Emotion Recognition, Facial Emotion Recognition, and Body Language Emotion Recognition, each produced by their respective modules described in earlier sections.
- Semantic & Structure: Sentence embeddings using SBERT [102], cosine similarity between sentence embeddings, coherence approximation, and topic drift detection.
- Temporal Analysis: Silence gap detection across the duration of the presentation.

In addition to the above, frame-level acoustic features are extracted, including loudness (decibel level per frame), pitch variation per frame, and the Signal-to-Noise Ratio (SNR) per frame. The Eye Contact Percent metric produced by the Eye Gaze module, described in the following subsection, is also incorporated into this same feature set. The output of Stage 1 is a structured metrics.json file, normalized and machine-readable, ready to be consumed by Stage 2.

Eye Gaze Model: A primary component of Stage 1 is the Eye Gaze tracking module, whose core objective is to compute an overall "Eye Contact Percent" metric across the duration of the video. The module relies on a combination of OpenCV [103] and MediaPipe Face Mesh [104, 105]. This stack was selected over alternative approaches for several strategic reasons. Deep learning-based gaze estimation models typically require dedicated training data, GPU acceleration, and considerably more inference time, making them difficult to justify for a sub-component whose only output is a single percentage metric. In contrast, the chosen approach relies on geometric reasoning over facial landmarks rather than a trained gaze estimation network, which keeps the module fast enough to run frame-by-frame on standard hardware without a measurable impact on overall pipeline latency.

The first major design decision was instantiating Media Pipe’s Face Mesh with the refine landmarks=True parameter. This setting enables high-precision iris localization by adding dedicated iris landmarks (468 and 473) directly to the standard face mesh output, removing the need for a second, independent iris-tracking neural network that would otherwise be required to achieve comparable precision. This single design choice significantly reduces the computational footprint of the module while still achieving sub-pixel iris localization accuracy.

The second major design decision was relying on mathematical spatial ratios rather than a black-box deep learning classifier to determine gaze direction. This decision was made for three reasons.

First, interpretability: a ratio-based threshold can be inspected, explained, and tuned manually, whereas a trained classifier's decision boundary cannot. Second, adjustability: the sensitivity thresholds (0.42–0.58 horizontal, 0.40–0.60 vertical) can be recalibrated for different camera angles or distances without retraining any model. Third, robustness to domain shift: since the method depends only on the geometric relationship between the iris and the eye boundary rather than on visual features learned from a specific training distribution, it generalizes naturally across different lighting conditions, camera qualities, and presenter appearances without requiring fine-tuning.

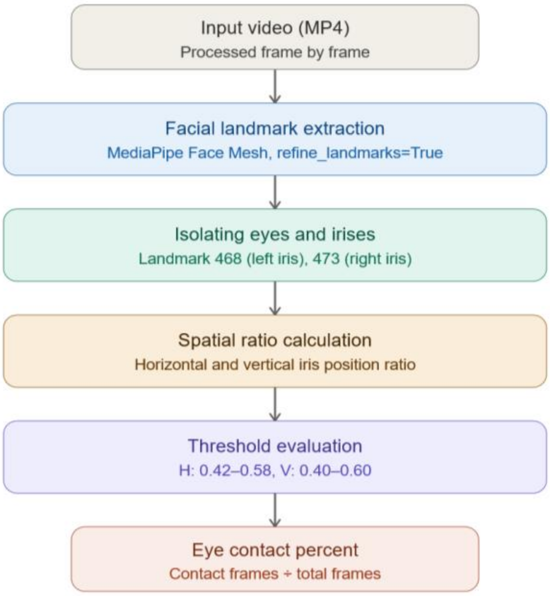

Figure 4. 17 Eye Gaze Pipeline.

The eye-tracking pipeline processes the input video frame by frame. MediaPipe Face Mesh detects a single face per frame and extracts 468+ facial landmarks, mapping their normalized coordinates to pixel locations based on the frame's width and height [104]. Explicit landmark index arrays are used to isolate the boundaries of the left and right eyes, while the center of the left iris (landmark 468) and right iris (landmark 473) are pinpointed directly. For each eye, a bounding box is computed from the minimum and maximum X and Y coordinates of the eye landmarks, and the position of the iris within this bounding box is expressed as a horizontal and vertical ratio, where 0.5 represents a perfectly centered iris. The left and right ratios are averaged to account for facial angle, and a frame is flagged as "looking at the camera" if the average horizontal ratio falls within 0.42–0.58 and the vertical ratio falls within 0.40–0.60. The final Eye Contact Percent is calculated as the number of frames flagged as maintaining eye contact divided by the total number of frames processed.

4.7.2 Analytical Reasoning Engine

Stage 2 acts as an objective, non-emotional data processor rather than a generic LLM prompt — it is designed as a controlled, data-driven reasoning system that minimizes emotional framing in its output, prioritizing factual, threshold-based assessment over subjective or affective language. It should be noted that large language models, including the one used in this stage, are trained on human-generated text and can therefore carry inherent biases from their training data [113]; this design choice constrains the system's output behavior rather than eliminating bias at the model level. It instantiates a single Gemini targeting the gemini-2.5-flash model [106], to evaluate speech and presentation performance.

Gemini 2.5 Flash was selected for this module, as generating structured analytical output from extracted performance metrics does not require the advanced reasoning, multimodal, or long-context capabilities offered by larger or more specialized models. Its text generation capabilities were sufficient to produce clear, well-structured analytical output from the metrics pipeline, and its free accessibility within our development environment further supported this choice, allowing iterative testing and refinement without incurring API costs.

This stage ingests the metrics.json file produced by Stage 1 alongside an external reference JSON file containing standard communication science thresholds that define the boundary between poor and excellent performance for each metric. The threshold values for each metric are derived from the empirical literature: acceptable WPM ranges follow research on vocal speed and audience persuasion [110], filler word rate cutoffs are informed by findings on disfluency and perceived speaking effectiveness [109], and vocal acoustic thresholds for pitch variation and loudness are drawn from the paralinguistic cue literature on speaker credibility [111].

Eye contact percentage thresholds follow research linking sustained gaze to perceived credibility in public speaking contexts [112]. Where the literature reports ranges rather than precise cutoffs, threshold values were placed at the conservative boundary of the reported effective range, so the system only flags deviations the existing evidence associates with a meaningful negative impact on audience perception. Because each input metric (such as pitch, WPM, and loudness) operates on a different numerical scale and carries a different meaning in the context of presentation quality, the Analysis Agent first assesses each value relative to its corresponding threshold — determining how far each metric deviates from acceptable performance — and then ranks metrics by severity of deviation to identify the most critical issues. It then identifies the top three critical presentation issues and formulates a highly structured summary using Observation-Impact-Suggestion (OIS) pairings, without drafting any long-form feedback text

The Analysis Agent returns a strict JSON schema, stripped of all markdown code block formatting, consisting of the top three issues (each with an issue name, severity, and category), a list of observations mapping each metric to its value, threshold, and status, a list of impacts describing the consequence of each issue, a list of short suggestion phrases, and a brief reasoning field explaining how issues were ranked.

Table 4. 6 Documented JSON Output Schema for Stage 2.

Field	Description
top_3_issues	Each entry contains the issue name, its severity level (e.g., high, medium, low), and its category.
observations	Each entry maps a metric name to its extracted value, its reference threshold, and its status (e.g., within range, below threshold).
impacts	Each entry pairs an issue name with a description of its consequence on presentation quality.
suggestions	Short, actionable improvement phrases corresponding to the identified issues.
reasoning	A brief explanation of how the top issues were ranked and how metrics were mapped to performance categories.

Advanced Behavioral Insights: The three behavioral detectors implemented in this stage were selected based on their grounding in well-established bodies of research on speaker perception. Each targets a cross-modal mismatch pattern that the communication science and affective computing literature identifies as a meaningful, observable indicator of presentation quality: emotional leakage between face and voice (a phenomenon documented in deception and masking research [114, 115]), body-verbal incongruence as a marker of speaking anxiety [116], and discourse coherence breakdown as a predictor of audience disengagement (motivated by the coherence and topic-drift signals established in Stage 1 and grounded in discourse processing literature). These were prioritized over other possible behavioral signals because each has an identifiable cross-modal detection mechanism, a corresponding literature base, and a direct link to how a presentation is perceived by an audience — consistent with the feature selection rationale applied in Stage 1.

Beyond raw metric thresholds, Stage 2 is designed to detect more complex behavioral patterns that emerge from mismatches between modalities. Reliably inferring a speaker's actual internal cognitive and emotional states from multimodal signals remains an open research problem in affective computing [19]; the detectors described below are therefore designed to surface observable cross-modal signal mismatches such as conflicting emotional cues across modalities rather than to claim direct access to a speaker's true internal state.

Each of the following behavioral detectors is implemented as a separate algorithmic component that computes cross-modal signal mismatches before passing results to the LLM, rather than relying on the LLM itself to detect these patterns from raw inputs. The resulting mismatch signals are also passed as raw inputs to Stage 3, where they may directly influence the generated feedback when relevant.

- Emotion Incongruence Detection: flags cases where a positive facial expression is paired with a negative vocal tone, labeled as "Masking / Forced Positivity." This draws on psychological literature on emotional masking, where an individual disguises a felt emotion by displaying a different one

[114], and on neuroscience research showing that mismatched facial and vocal emotional cues are detectable as a distinct form of cross-modal conflict [115].

- Confidence Gap Detection*:* flags cases where positive speech content is paired with a tense body posture, labeled "Hidden Nervousness." This is grounded in research linking rigid, closed body posture to speaking anxiety, even when verbal content does not explicitly convey distress [116].

- Discourse Structure Modeling: identifies points in the presentation where logical flow likely breaks down, based on the sentence-level coherence and topic drift signals computed in Stage 1. A sustained drop in semantic similarity between consecutive sentences, or a sharp shift in topic relative to the surrounding context, is interpreted as a potential discourse break — a point where the presentation's logical thread weakens and audience attention is more likely to be lost.

4.7.3 Coaching and Feedback Agent

The final stage acts as a public speaking coach, mapping the analytical scores.json data produced by Stage 2 into human-readable text that is warm, specific, and motivational. This stage follows three core structure rules. First, the Sandwich Method is applied: the report begins with genuine praise, flows into constructive OIS breakdowns, and concludes with motivational encouragement [107]. Second, OIS Delivery Execution ensures that every core issue lists a direct observation drawn from the data, its psychological impact on an audience, and one concrete, actionable improvement phrase [108].

Third, a set of strict boundaries is enforced: only the top three issues are evaluated, emojis are never used, and the tone is kept supportive and tailored for an anxious but eager student. The agent is explicitly restricted from contradicting any analytical data provided by Stage 2, ensuring consistency between the objective scoring and the delivered feedback.

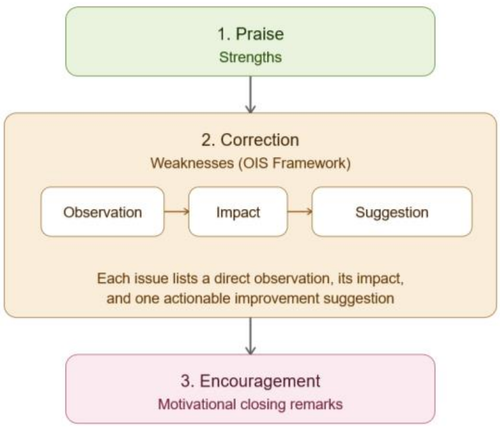

Figure 4. 18 Report Structure diagram based on sandwich method [107].

The final output is wrapped in a clean JSON layout designed to safely pass formatted arrays downstream to the frontend UI. It consists of a praise field containing genuine positive highlights, a constructive feedback array where each entry pairs an issue name with its complete OIS narrative text, an encouragement field containing motivational closing remarks, and a full_formatted_text field containing the complete combined compilation of all sections.

Table 4. 7 Documented Output Schema for Stage 3.

Field	Description
`praise`	Genuine positive highlights drawn from the user's presentation performance.
`constructive_feedback`	Each entry contains an issue name paired with its complete Observation-Impact-Suggestion (OIS) narrative text.
`encouragement`	Motivational closing remarks tailored to the user's delivery.
`full_formatted_text`	A complete combined compilation of the praise, constructive feedback, and encouragement sections, formatted for direct display in the frontend UI.

4.7.4 Emotions Integration

Integrating multimodal emotion tracking body emotion, facial emotion, and vocal emotion extends the coaching system from a basic metrics checker into a more intuitive, human-like presentation mentor [100]. Emotion signals are incorporated into the feedback pipeline in four main ways. Emotional Congruency Scores assess whether facial, body, and speech emotions are mutually consistent. For instance, if the speech emotion indicates high energy or excitement while facial and body expressions indicate anxiety or flatness, this mismatch is flagged as a "Delivery Incongruency" metric.

Confidence vs. Anxiety Indices group related emotional markers into aggregated behavioral states, following the dimensional approach used in affective science, where discrete emotions are combined into composite indices of broader affective states (e.g., positive/negative affect) rather than analyzed in isolation [118]. The current implementation groups emotions across two cross-modal pairings facial and speech emotions, and body and facial emotions checking whether the emotions within each pairing are mutually consistent. Repeated occurrences of specific emotions can carry different meanings depending on their combination and context, and these groupings remain under active refinement as the system develops.

Emotional Trajectory tracks how emotional states shift across the timeline of the presentation. This is informed by the serial position effect, a well-established finding in memory research showing that information presented at the beginning and end of a sequence is retained more strongly than information in the middle [118] suggesting that a presentation's opening and closing moments may carry disproportionate weight in shaping audience perception. Building on this, the system treats a stable emotional baseline with strong opening and closing engagement as a favorable pattern to detect erratic emotional fluctuations, the system identifies the most frequently occurring emotion across the presentation as the baseline emotional state, then counts the number of frames where the detected emotion deviates significantly from this baseline — these deviations are treated as emotional spikes. The total spike count is passed as an input to the LLM, which interprets it as a potential indicator of reduced composure if it exceeds the expected range for a well-controlled presentation.

Emotional signals are integrated at two stages of the pipeline. In Stage 2, they are passed as raw analytical inputs alongside the metric data, allowing the Analysis Agent to factor emotional patterns into its objective assessment. In Stage 3, the Feedback Coach incorporates these signals primarily within the encouragement section and closing remarks, grounding the coaching language in the speaker's actual emotional delivery rather than relying on text-derived metrics alone. The core OIS breakdowns remain driven by the objective metric analysis from Stage 2, ensuring that constructive feedback is grounded in measurable data rather than inferred emotional state.

4.8 Testing and Experimental Results

This section documents the experimental trials conducted for each module, summarizing the configurations explored and the performance achieved across successive runs. For each module, experiments were designed to isolate the impact of specific design decisions such as training strategy, preprocessing choices, feature extraction parameters, and architecture variants and compare their effect on the final evaluation metrics. The results presented in the following tables reflect an iterative process in which each trial builds on lessons learned from the previous one. The goal was not simply to maximize accuracy on a single benchmark, but to identify configurations that generalize well and align with the specific constraints and requirements of each module within the Elevare pipeline.

4.8.1 Question Generation Experiments

Table 4.8 presents the empirical evaluation metrics for the Question Generation (QG) module across different datasets and scale configurations utilizing a structured few-shot prompting strategy. The experimental setup employed a 6-shot prompting technique augmented with expert-validated examples, boundary rules, and answer anchor enforcement.

Initial testing on a smaller subset of the CoQA dev set (16 stories, 96 QA pairs) yielded the highest performance, achieving an Answerability F1 score of 0.906, a Paragraph Score (Para Score) of 0.664, and a robust Bloom Alignment of 92.71%. As the dataset scale expanded to 100 stories (800 QA pairs) on the CoQA dev set, a minor degradation in performance was observed, with the Answerability F1 decreasing to $.856 and Bloom Alignment shifting to 84.75%. This minor drop indicates the typical scalability challenges and behavioral variance when LLMs process a broader array of contextual structures.

Table 4. 8 QG Experimental Results.

Data	# Stories	# QA pairs	Training Technique	QG metrics		Bloom metrics	
				Answerability F1	Para Score	Bloom Alignment	Anchor Presence
CoQA dev set	16	96	● 6-shot prompting with expert-validated examples, boundary rules ● Answer anchor enforcement	0.906	0.664	92.71%	98.96%
CoQA dev set	100	800		0.856	0.654	84.75%	97.5%
QG (Source from CoQA)	100	640		0.628	0.600	85.78%	97.03%

Furthermore, when testing the pipeline on the QG-specific subset sourced from CoQA (100 stories, 640 QA pairs), the system maintained solid structural stability, securing a Bloom Alignment of 85.78% and an exceptionally high Anchor Presence of 97.03%, despite a drop in the Answerability F1 score to 0.628. Across all three experimental configurations, the Anchor Presence consistently remained remarkably high (ranging from 97.03% to 98.96%). This stability demonstrates the absolute effectiveness of the enforced anchor constraints in ensuring that the generated conversational QA pairs remain strictly grounded in the reference source text, effectively mitigating hallucination risks

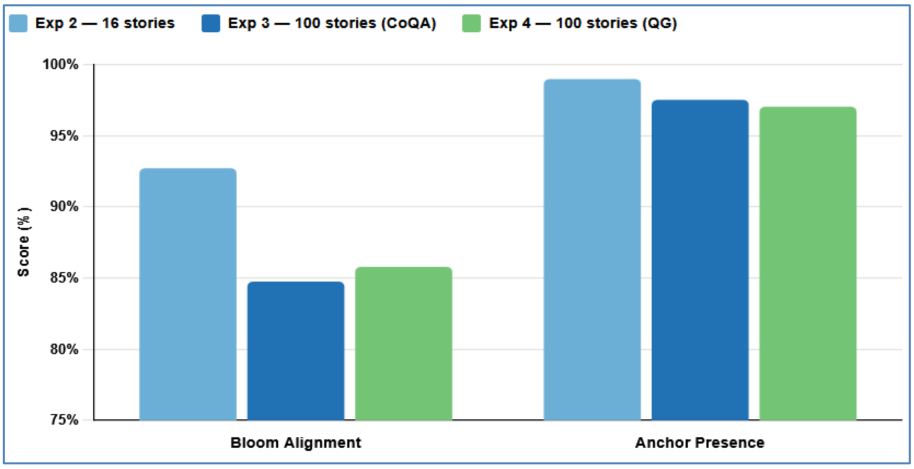

Figure 4. 19 Bloom Alignment and Anchor Presence remain stable across Exp 2⟶3 (scaling) and Exp 3⟶4 (dataset switch), with all metric shifts within 8 percentage points.

4.8.2 Question Answer Validation Experiments

The evaluation results presented in the table demonstrate the performance of the evaluation framework on the Saf_communication_networks_english dataset, comparing two distinct prompting paradigms across an identical multi-component output layer. The output configuration required the language model to simultaneously calculate a granular quantitative score (0–100), assign a strict qualitative category (Correct, Partially Correct, or Incorrect), and synthesize structured actionable feedback.

Table 4. 9 Question Answer Validation Results

Dataset	Training Technique	Output Layer	Accuracy
Saf_communication_networks_english	Zero-Shot using DSPy Chain Of Thought	• Calculates a final score (0-100) • Strict label (Correct, Partially Correct, Incorrect) • Actionable Feedback	72.48%
	Zero-Shot using Manual prompt engineering		72.26%

When leveraging programmatic optimization via Zero-Shot using DSPy Chain of Thought (CoT), the validation framework achieved its highest performance with a grading accuracy of 72.48%. In comparison, relying on Zero-Shot using Manual prompt engineering yielded a slightly lower accuracy of 72.26%.

Although the numerical delta between the two techniques is relatively narrow (0.22%), these findings underscore the empirical value of structured, algorithmic prompt optimization. The marginal superiority of the DSPy framework indicates that decomposing the evaluation process into explicit step-by-step reasoning steps through a systematic Chain-of-Thought pipeline enhances structural stability and reasoning alignment, allowing the model to better navigate the complex task of multi-criteria answer validation without requiring explicit few-shot exemplars.

4.8.3 Facial Emotion Recognition Experiments

Table 4.10 demonstrates the empirical progression of fine-tuning a ConvNeXt architecture augmented with a custom Detail Extraction Block on the FERPlus dataset, compared against the baseline metrics reported in the literature. While the reference paper architecture achieves an optimal Test Accuracy of 95.69%, initial experimental runs highlighted the critical role of training strategies and architectural freezing in overcoming underperformance. In Trial 1, using a basic preprocessing pipeline and a learning rate of 0.01, the model struggled, yielding a Test Accuracy of 63.56% and a Balanced Accuracy of 56.93%. Introducing heavy geometric augmentation during the remaining epochs in Trial 2 provided a marginal improvement, raising the Test Accuracy to 67.53%.

However, the most notable optimization occurred in Trial 3, where the introduction of a dual-stage training technique alongside unfreezing the last two stages of the backbone architecture, significantly enhanced feature discrimination. This configuration pushed the Test Accuracy to 68.50% and maximized the Balanced Accuracy to 62%. Despite these incremental enhancements achieved through targeted optimization, the persistent performance gap relative to the original paper's benchmark provided the empirical justification for the subsequent decision to pursue the GiMeFive convolutional network as the primary architecture for the facial expression recognition module.

Table 4. 10 ConvNeXt + Detail Extraction Block results on FERPlus dataset:

<table>
<tr><th>Trials</th><th>Pre-Processing</th><th>Training technique updates</th><th>Architecture updates</th><th>Results</th></tr>
<tr><td>Paper Results</td><td rowspan="2">Resize 224x224
Random H-Flip
Normalization</td><td>Lr = 0.001</td><td rowspan="3">NA</td><td>Test Acc.:95.69%</td></tr>
<tr><td>Trial 1</td><td rowspan="2">Lr= 0.01</td><td>Test Acc.: 63.56%
Balanced Acc:56.93%</td></tr>
<tr><td>Trial 2</td><td>First 5 epochs: same as run 1
Remaining epochs: heavy geometric augmentation</td><td>Test Acc.: 67.53%
Balanced Acc.: 58.68%</td></tr>
<tr><td>Trial 3</td><td>Resize 224x224
Random H-Flip
Normalization</td><td>Stage 1: Vanilla warm-up
Stage 2: 16 epoch, label-distribution-aware margin + Deferred Reweighting (1/√n weights)</td><td>Backbone last 2 Stages unfrozen in Stage 2</td><td>Test Acc.: 68.50%
Balanced Acc.: 62%</td></tr>
</table>

The decision to pursue GiMeFive as the primary architecture for the facial emotion recognition module was driven by the nature of the RAF-DB dataset. RAF-DB presents a severe class imbalance, with the Happiness class heavily dominating and minority classes such as Fear and Disgust being significantly underrepresented. This imbalance made it unsuitable to train the ConvNeXt + DEB model, which had been designed and tuned for FERPlus — a different distribution — without substantial rearchitecting of the imbalance-handling pipeline.

A review of the literature identified GiMeFive [10] as a model explicitly validated on RAF-DB under comparable conditions, with preprocessing and augmentation strategies directly compatible with the dataset's structure. Its lightweight design also allowed more aggressive class-aware augmentation and deferred re-weighting strategies to be applied without excessive computational overhead. Accordingly, GiMeFive was adopted as the final model for this module, with the experimental runs in Table 4.11 documenting the progressive improvements achieved through targeted imbalance mitigation.

Table 4. 11 GiMeFive results on RAF-DB dataset

Trials	Pre-Processing	Training technique updates	Results
Paper Results	• Resize 64x64 • Normalization only (no random augmentation) • RAF-DB aligned & cropped images	Standard Cross Entropy loss . SGD lr=0.001 .Dropout + BN No squeeze-excitation	Test Acc.: 86.5%
Trial 1	• Resize 64x64 • Minority: heavy augmentation • Majority: light augmentation (flip only)	Standard CE loss SGD lr=0.001 . warm restarts	Test Acc.: 83.5%
Trial 2	• Surgical augmentation adjustments	Phase 1: Plain Cross Entropy (ep. 1-80) Phase 2: Deferred Re-weighting with sqrt class weights + Mixup	Test Acc.: 84.09% Macro F1: 73.80% Macro F1 Recall: 73.44%
Trial 3	• Majority: light augmentation • Minority: heavy augmentation-based oversampling	Same as run 1	Test Acc.: 84.46% Macro F1: 73.45% Macro F1 Recall: 70.63%

4.8.4 Speech Emotion Recognition Experiments

Section 4.3.4 presents the evaluation of the Speech Emotion Recognition module across different configurations. While the reference paper run utilizing a 5-fold cross-validation achieved a peak Test Accuracy of 94.03% and a Macro F1 of 0.94, our internal empirical trials adopted a strict 80%-20% train/test split.

Within these runs, optimizing the Principal Component Analysis (PCA) dimensions proved critical; increasing PCA components from 90 (Trial 1) to 110 (Trial 2) enhanced the Test Accuracy from 90.4% to 91.4%. Ultimately, Trial 3 balanced this trade-off with 97 PCA components, yielding a stable 92.9% accuracy and a high Macro F1 score of 0.933, confirming the robustness of the extensive audio preprocessing and feature extraction pipeline.

Table 4. 12 Speech Emotions Results

Trials	Preprocessing	Feature Extraction	Training Technique	PCA	Test Accuracy	Macro F1
Paper run	• Silence Removal: 70% removed if > 200ms • Resampling: all audio standardized to 22050 Hz • Gaussian noise injection: rates 0.035 / 0.025 / 0.015 • Pitch shifting: rates 0.7 / 0.8 / 0.7 (no duration change) • Time stretching: factors 0.8 / 0.9 / 0.7 (no pitch change)	• Mel-Frequency Cepstral Coefficients (MFCCs • Including Delta (first derivative) Delta-Delta (second derivative), and standard deviation • Chroma-Based spectral features • Logarithmic Mel-Scaled Spectrogram • Spectral Contrast Features • Root Mean Square Energy (RMSE) • Zero Crossing Rate (ZCR)	5 fold cross validation	97	94.03%	0.94
Trial 1			train/test split 80%-20%	90	90.4%	0.908
Trial 2				110	91.4%	0.918
Trial 3				97	92.9%	0.933

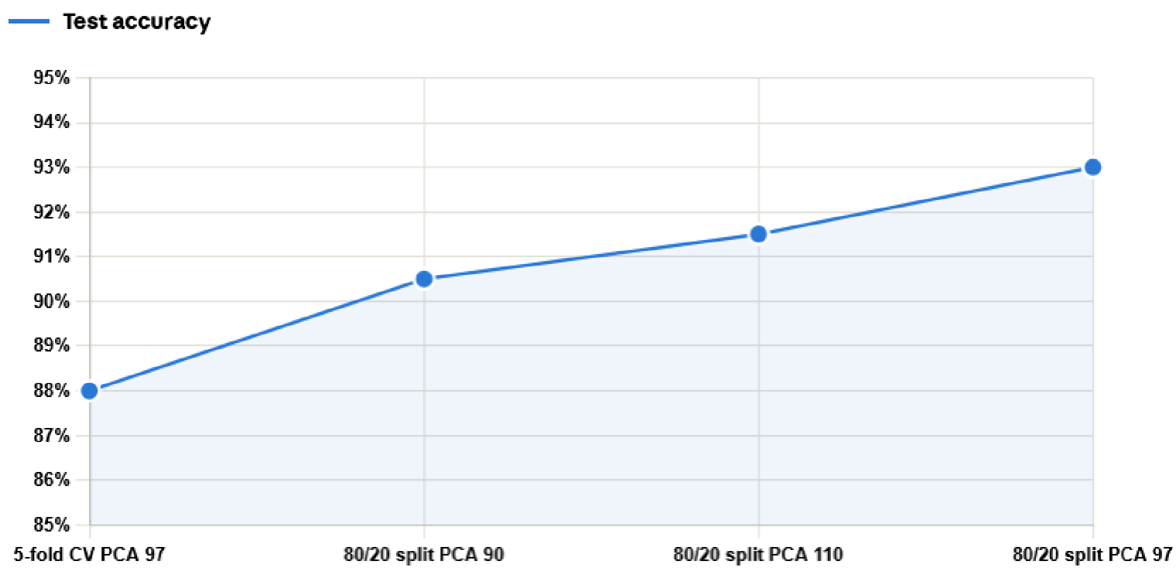

Figure 4. 20 Test accuracy across evaluation configurations, peaking at 93% with an 80/20 split and PCA 97.

4.8.5 Body Language Emotion Recognition Experiments

The experimental results evaluate a posture classification baseline utilizing a shared preprocessing and distance-based feature extraction pipeline. Under a subject-dependent testing technique (Trial 1), where the 80%-20% train/test split includes data from the same individuals across both subsets, the model achieves an accuracy of 69%. However, transitioning to a more rigorous subject-independent validation (Trial 2) causes a significant performance drop to 52.4%. This performance degradation underscores the challenging nature of cross-subject generalization, highlighting the model's sensitivity to individual anatomical and movement variations when evaluating entirely unseen subjects.

Table 4. 13 KNN Experimental Results.

Model	Preprocessing	Feature Extraction	Testing Technique	Accuracy
Trial 1	• Confidence Filtering: Remove joints with confidence < 0.5 • Centering • Scale Normalization • Flatten normalized (x,y) -> 50 features • Class Balancing	Distance	Train/Test split 80%-20% (subject-dependant)	69%
Trial 2			Train/Test split 80%-20% (subject-independant)	52.4%

Table 4. 14 SVM Experimental Results.

Model	Preprocessing	Feature Extraction	Testing Technique	Accuracy
Trial 1	• Confidence Filtering: Remove joints with confidence < 0.5 • Centering • Scale Normalization • Flatten normalized (x,y) -> 50 features • Class Balancing	Distance	Train/Test split 80%-20% (subject-dependant)	69%
Trial 2			Train/Test split 80%-20% (subject-independant)	52.5%

Table 4.15 presents the empirical results of the Spatial-Temporal Transformer (ST-TR) model, demonstrating contrasting performance based on the evaluation technique used despite sharing identical preprocessing and feature extraction pipelines (velocity, acceleration, and body joints). Under a subject-dependent testing framework (Trial 1), the model achieves a classification accuracy of. In contrast, transitioning to a rigorous subject-independent validation (Trial 2) causes a significant performance drop to, highlighting the substantial challenge the transformer architecture faces in generalizing across unseen individuals due to natural variations in body anatomy and motion dynamics.

Table 4. 15 ST-TR Experimental Results.

Model	Preprocessing	Feature Extraction	Testing Technique	Accuracy
Trial 1	• Key point interpolation • Centering • Scale Normalization • Rotation Normalization • Sequence Length Standardization • Join Subset Selection • Class Balancing	• Velocity • Acceleration • Body joints	Train/Test split 80%-20% (subject-dependant)	73%
Trial 2			Train/Test split 80%-20% (subject-independant)	56.3%

Table 4. 16 MLP Experimental Results

<table>
<tr><th>Trials</th><th>Preprocessing</th><th>Feature Extraction</th><th>Flattening Technique</th><th>Testing Technique</th><th>Best Accuracy</th></tr>
<tr><td>Trial 1</td><td rowspan="6">Missing Key point interpolation
Centering
Scale Normalization
Sequence Length Standardization
Class Balancing
Missing Key point interpolation
Centering
Scale Normalization
Sequence Length Standardization
Class Balancing</td><td rowspan="6">Distance
Angle
Area
Velocity
Acceleration
Distance
Angle
Area
Velocity
Acceleration</td><td rowspan="6">Mean
Standard deviation
Median
Range
Energy
Mean
Standard deviation
Median
Range
Energy</td><td>Train/Test split 80%-20% (subject-dependant)</td><td>79.7%</td></tr>
<tr><td>Trial 2 (architecture tuning)</td><td>Train/Test split 80%-20% (subject-independant)</td><td>83.2%</td></tr>
<tr><td>Trial 3</td><td rowspan="3">Train/Test split 75%-25% (subject-dependant)</td><td>73%</td></tr>
<tr><td>Trial 4</td><td>77%</td></tr>
<tr><td>Trial 5</td><td>73%</td></tr>
<tr><td>Trial 6</td><td>Train/Test split 75%-25% (subject-independant)</td><td>69.3%</td></tr>
</table>

The body language experiments followed a clear progressive trajectory, with each model addressing limitations identified in the previous one. The classical baselines — KNN and SVM — established an initial performance ceiling using handcrafted distance-based features, achieving acceptable results under subject-dependent evaluation but dropping significantly under subject-independent splits, revealing limited generalization. The ST-TR model introduced a more expressive architecture capable of capturing spatial and temporal dependencies across joints simultaneously, improving subject-independent performance, but its design assumptions — originally built for action recognition — did not fully align with the subtler, holistic nature of emotional body expression.

The MLP addressed this by incorporating a richer set of geometric and kinematic features, achieving the strongest results among the non-sequential approaches. However, collapsing the temporal dimension into statistical aggregates discarded motion dynamics that are critical for emotion recognition. The Temporal CNN was therefore adopted as the final architecture, explicitly modeling multi-scale temporal dependencies across three parallel feature streams — joint positions, bone vectors, and velocity — while retaining the geometric richness of the MLP's feature set. This combination yielded the most robust and generalizable performance across the evaluated configurations.

Chapter 5
Conclusion
& Future Work

Chapter 5: Conclusion and Future Work

5.1 Conclusion

This project addresses the widespread challenges of public speaking anxiety and the limitations of traditional, subjective human feedback. Effective communication is a critical soft skill, yet many individuals struggle with imprecise delivery, poor non-verbal cues, and an inability to handle spontaneous questions. While several commercial AI speech coaches exist, they predominantly focus on audio-linguistic metrics (such as filler words and pacing) and fail to capture the holistic physical dimensions of a presentation. Furthermore, existing systems lack the contextual awareness required to simulate a high-stakes interactive panel, leaving a significant gap in comprehensive presentation coaching.

To bridge this gap, we developed Elevare, an AI-powered multimodal public speaking coach and presentation simulator. The system successfully evaluates speakers across a complete spectrum of communication channels. By integrating advanced deep learning architectures, the system achieved robust classification accuracies across its core modules: a Stacked BiLSTM network for Speech Emotion Recognition (92.90%), an optimized GiMeFive architecture for Facial Emotion Recognition (84.46%), and a highly robust Temporal CNN for Spatial-Temporal Body Language tracking (89.5%).

Beyond physical and acoustic tracking, Elevare introduces a highly innovative cognitive evaluation pipeline. The system utilizes Semantic Graph Conversational Question Generation (SG-CQG) to automatically ingest a user's presentation slides and generate contextually relevant questions. The user's spontaneous responses are then evaluated using a zero-shot retrieval-augmented validation module driven by DSPy Chain-of-Thought prompting and Llama 3.3-70b, achieving a strict grading accuracy of 72.48%.

Finally, the project adds immense value through its Dual-Agent Feedback Architecture. By strictly decoupling the objective mathematical reasoning engine from the empathetic coaching language model, Elevare prevents hallucination and emotional bias from skewing the performance metrics. This ensures that users receive actionable, precise, and scientifically grounded pedagogical feedback. Ultimately, Elevare transitions automated speech tracking into a holistic, interactive presentation simulator, empowering users to improve both their physical delivery and their contextual mastery of the material.

5.2 Future Work

To build upon the current capabilities of the Elevare platform, several critical avenues for future research and development will be pursued. First, establishing real-world academic integration through active collaborations with educational institutions will provide invaluable, live-environment empirical testing. By deploying the system in actual university seminars and gathering user-experience data from students actively preparing for high-stakes graduation project defenses, the pedagogical tone and contextual behavior of the Dual-Agent feedback loop can be iteratively refined based on genuine academic stressors.

Furthermore, expanding the system's linguistic capabilities to include multilingual support represents a key evolutionary step. While currently optimized for English, extending the natural language processing, conversational question generation, and speech emotion recognition modules to support Arabic would drastically enhance regional accessibility. This localization would allow the platform to cater directly and more effectively to the distinct communicative nuances of the local educational and corporate sectors.

Finally, to bridge the gap between automated coaching and true situational immersion, future work will explore the incorporation of immersive audience simulation via virtual avatars. Integrating interactive, human-like avatars capable of reacting in real time to the speaker's delivery—such as exhibiting non-verbal signs of disengagement, nodding in agreement, or generating spontaneous verbal interruptions—will accurately replicate the complex psychological pressures of live public speaking. This dynamic environmental feedback will effectively condition users to manage stage fright, interpret room dynamics, and adapt their delivery under realistic conversational stressors.

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